River Deep Mountain AI

D2.3 – Insights report from stakeholder research

WP2: Stakeholder Engagement and Outreach

River Deep Mountain AI

Version overview

Version	Date	Authors	Comments
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Sign-off

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1 Overview

This document outlines the key findings from Phase 1 of the River Deep Mountain AI (RDMAI)project, specifically WP2 stakeholder engagement.

The aim of the stakeholder engagement was to identify critical drivers for enhancing waterbody health and clarify where AI/ML can be most impactful. Contributions from leading experts across environmental NGOs, water companies, regulatory bodies, AI, charities and academia helped validate and refine the initial use-cases from the original project submission. The findings are presented through focused themes and building blocks.

Stakeholder Engagement Process and Definitions

Before RDMAI formally started, a range of ideas for models or water body health challenges that the project could focus on were suggested. These ideas came from *ad hoc* conversations, virtual workshops, and a three-day workshop at Northumbrian Water's 2024 Innovation Festival. The ideas included predicting algal blooms, finding misconnected sewers and using satellite imagery to better understand how increasing the carbon capture potential of soils could reduce the amount of runoff (and thereby pollution) that those soils generate – all ideas are described in the RDMAI IF 2024 wrap-up .

The ideas were refined through the initial 'discovery and definition' phase of RDMAI (August to October 2024). Structured stakeholder research was undertaken to generate a shortlist of priority 'themes' – i.e. topics on which RDMAI would focus. The themes covered some ideas that were suggested before the project started, but also generated new ideas and led to some ideas not being considered further. This report details the structured stakeholder research, findings, and themes identified.

The terms referred to in this report are outlined below, and the relationship between the terms is illustrated by Figure 1.

Theme – *An overarching topic that has been identified as being of interest to stakeholders.*

Building Block – *The defined themes are broad in nature; to address them effectively, they are broken down into more specific, subcategories termed 'AI/ML building blocks'. The building blocks within each of the themes detail the ways in which each theme could be 'modelled'. The AI/ML building blocks represent tangible goals aligned with each theme and serve to focus efforts on distinct, solvable challenges. The AI/ML building blocks are essentially modelling approaches, that help highlight the ways in which a theme can be addressed.*

Use case – *Outlines how a stakeholder could use model outputs. A use case might be made up of more than one building block but should fit within at least one of the themes.*

Artificial intelligence (AI) – Artificial intelligence is a field of science concerned with building computers and machines that can reason, learn, and act in such a way that would normally require human intelligence or that involves data whose scale exceeds what humans can analyze[1]. ML is a focused branch of AI, where systems improve their accuracy through data exposure rather than explicit programming for each scenario. Trained on vast datasets, ML models detect patterns, predict outcomes and classify information.

Modelling – Throughout this report, we refer to AI/ML, models and modelling. Unless otherwise specified, all these refer to the training and development of artificial intelligence using different techniques from the field of AI, including machine learning, such as deep learning, neural networks and more.

Figure 1: Taxonomy of themes, building blocks and use cases.

[1] https://cloud.google.com/learn/what-is-artificial-intelligence

Structured Stakeholder Research

66 stakeholders were invited to an interview from 25 organisations. Through the interviews, we aimed to ensure that River Deep Mountain AI model development focused on the themes that would best address stakeholder needs. Specifically, the interviews aimed to:

- Understand the actions that stakeholders take to improve the health of water bodies. What are the advantages and disadvantages of current approaches to understanding water body health problems?
- Understand who our model end users are – who do model outputs need to be suitable for? Are there existing collaborations/networks that we could work with or expand on during the next phases of the project?
- Understand data quality and availability – what data is available, how is it used currently, and how could RDMAI access the data?
- Identify potential pilot catchments for model development and testing.
- Gather information on relevant previous projects – including models, academic papers, data collection, modelling, river restoration and similar projects.
- Learn about previous applications of AI to environmental challenges.
- Understand stakeholders' views on AI and machine learning. What are their fears and hopes? How could RDMAI help their work?

Interviewees that would help to meet the aims of the stakeholder research outlined above were identified. 206 potential contacts were found using the RDMAI project team's existing networks, from internet searches and from individuals with known interest in RDMAI (e.g. people who signed up to Northumbrian Water's 2024 Innovation Festival). From this list, 66 people from 25 organisations who would best help to meet the stakeholder research aims were invited to an interview. Figure 2 shows the organisation types that these 66 stakeholders represented. We aimed to ensure that a wide range groups impacted on or impacted by water body health were included. 42 stakeholders responded and were interviewed from August to October 2024 inclusive (dark blue bars in Figure 2). Figure 3 shows the organisations represented by the interviewees.

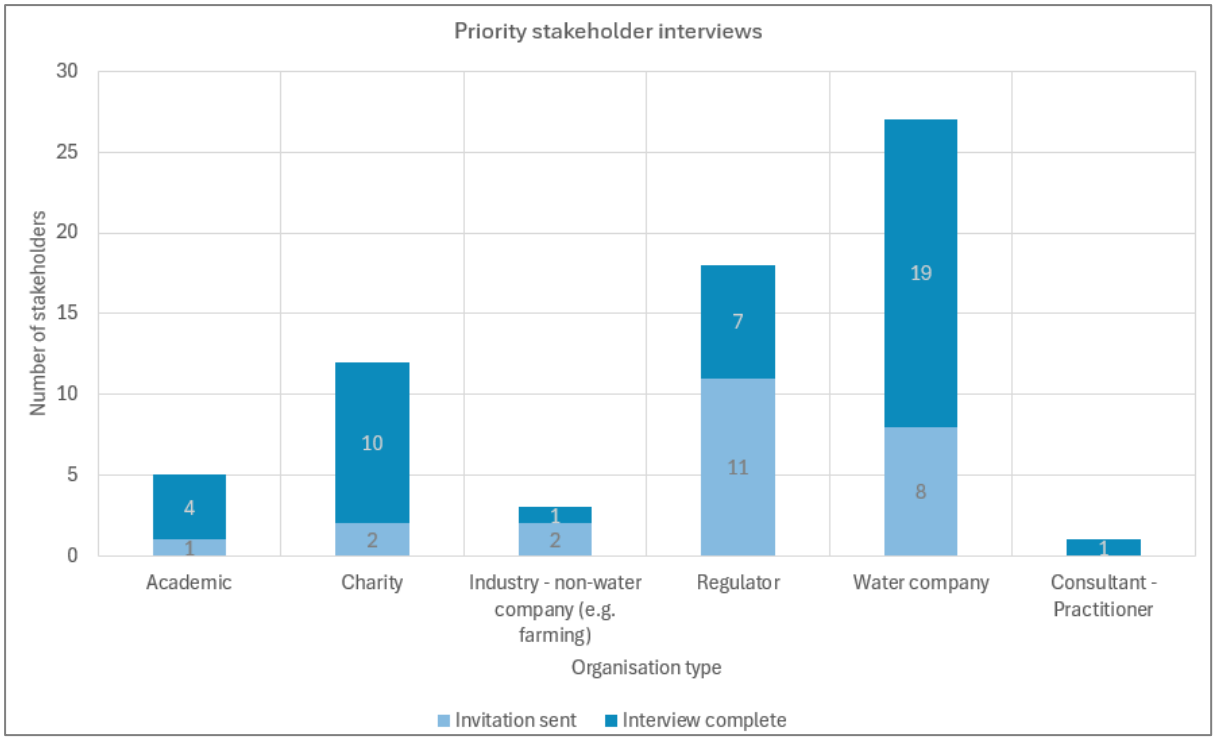

Figure 2: Organisation types represented during the interviews. Note that many of the charity employees interviewed were practitioners and data science specialists.

Figure 3: Organisations represented by the interviewees.

Interview structure

The interviews were semi-structured and followed interview guides developed for the project for either technical specialists (covering data scientists and academics) or practitioners (water companies, charities, industry and regulators).

Key points from the interviews were summarised by the RDMAI project team and used to define eight overarching themes – i.e. priority areas that model development should focus on. These themes are detailed below. When relevant, interviewees are referenced specifying if an inputs comes from a water company (W), charity (C), regulator (R), industry (I), academic (A) or other (O).

The interviews also highlighted potential pilot catchments where models could be developed and tested. These pilot catchments were linked to the themes and are also outlined below.

Where interviewees commented on data sets that RDMAI could draw on, these were added to the RDMAI project data register for further exploration by the project team.

The interviewees further outlined principles that are applicable to all themes and models developed through RDMAI. These included:

- **Enabling collaboration** – A holistic, catchment-based approach that brings together stakeholders impacting on and impacted by water body health is important. Model outputs should support equitable investment in our water bodies, and not focus solely on one industry or polluter.
- **Effective communication** of model outputs to different stakeholders will be critical to ensure their long-term use and development.
- **Model uncertainty** needs to be robustly understood and clearly explained to model users.
- **Build from previous work** – There are a lot of examples of AI being applied to environmental challenges, and a lot of work has already been done to understand and improve the health of our water bodies. RDMAI needs to build on this.
- **Transparency** - Peer review of the models and clearly communicating how they have been built will be vital to build trust and ensure long-term benefits from RDMAI.

Findings and opportunities

Overarching themes

Eight overarching themes were identified through stakeholder engagement interviews:

1. **Modelling high priority pollutants**
   Monitoring is expensive and testing for high-priority pollutants such as E. coli and Phosphorous is particularly challenging. An approach to overcoming this is to use the data that is already being collected in a better way.

2. **Designing an optimised monitoring plan**
   Existing datasets often lack the accuracy and reliability to adequately inform decision processes. Understanding where and what to monitor within a catchment can be challenging and expensive to work out.

3. **Delivering more value from continuous monitoring**
   The Environment Act requires English water companies to continuously monitor upstream and downstream of their assets. This creates unprecedented amounts of incoming data and related challenges.

4. **Monitoring biodiversity – species and habitat**
   Biodiversity is a key indicator of waterbody health. Measurements and changes over time are used a proxy of waterbody health across the sector and can be used as a way of measuring impacts of pollution.

5. **Understanding the impact of interventions to improve water body health**
   Due to the heterogeneous nature of historical solutions, along with the lack of targeted water quality monitoring post-implementation, it has remained difficult to quantify the effectiveness of these different interventions.

6. **Planning drinking water abstraction**
   The process of abstracting and treating drinking water is constrained by changes in waterbody health. Pesticides, nutrients, algae blooms, taste, odour and bacteria are all difficult to monitor and plan for. Reducing pollution upstream is the main need for drinking water stakeholders.

7. **Estimating the presence of 'chemicals' and emerging pollutants**
   Chemicals, heavy metals and other pollutants such as PFAS, microplastics, road runoff, PAHs, and more have a significant impact on waterbody health but are difficult to monitor. Having a better picture of emerging pollutants will increase the ability to improve waterbody health.

8. **Estimating the impacts of climate change on water body health**
   Climate change is expected to accelerate waterbody health challenges. Droughts and heavy rainfall events are expected to increase, making water management increasingly challenges. The impacts of this are difficult to predict and prepare for.

Theme prioritisation

To prioritise modelling during the next phases of River Deep Mountain AI, we evaluated which of the eight themes were likely to have the greatest impact on water body health, and the feasibility of developing AI models that could address each theme. Those themes with the highest potential impact and feasibility were selected to progress to the next phase of the project – model development and testing.

The evaluation followed the 'Theme screening - level 1' framework detailed in Table 1. This framework was designed to be quick to apply and to limit subjectivity; we aimed to ensure that the decision about which themes progress was robust and defensible. The framework is based on a similar framework developed for the Ofwat-funded Stream project. It was reviewed by the RDMAI Technical Advisory Group and by project partners from Northumbrian Water, WRc, Xylem and Cognizant. Scores were assigned to each theme by WRc, and then reviewed by funded project partners.

The total impact and feasibility scores for each theme are shown in Figure 4. Due to budget and programme constraints, and in line with the proposal that received funding from Ofwat's Innovation Fund, all eight themes could not progress through the project. Following the screening, theme 1 - modelling to understand high priority pollutants and theme 2 - designing an optimised monitoring plan were fast-tracked to the next phase of the project. Given their high potential impact and feasibility of developing models, they will be a priority during the model development phases of the project. Theme 3 – delivering more value from continuous monitoring – will also be progressed through the project, albeit with slightly lower priority than themes 1 and 2. Themes 4 and 5 have potentially high impact and/or feasibility but require further investigation to confirm value and feasibility of developing models. Based on resources (staff skills and availability) and project timescales, theme 4 progresses with the lowest priority and will be addressed only if time allows during phase 2. Themes 6, 7 and 8 may be a focus of future projects, but will not be progressed further through RDMAI.

Full scoring details for each theme can be found here.

As indicated by Table 1, further investigations into the feasibility of developing models for each of the prioritised themes will be undertaken during the next phases

of the project (see ‘Theme screening – level 2’). This will be based on deeper investigations into data availability and model testing.

Table 1: Theme screening framework - how did we choose which themes to develop models for?

		Score:	High (3)	Medium (2)	Low (1)	N/A (0)
			Theme screening – level 1			
Impact	**Could AI offer benefits over current approaches?**		Many complex, quantitative inputs that need condensing, AI offers speed and efficiency over current approaches, or clearly defined AI method that could be applied.		Resources required to create/maintain the model outweigh the outputs - e.g. extensive training would be needed if wanted to transfer model to another catchment.	
	Scalability	How transferable could the model(s) be? Which regions could they be applicable to?	UK and Ireland	Single region in UK and Ireland (e.g. focus for England only).	Limited no. of UK and Ireland catchments	
	People	How many stakeholder groups or organisations referenced this theme in interviews and could directly benefit from model outputs? Stakeholders should be different types - e.g. if 10 water companies mentioned this, only count once.	4+ groups	3-4 groups	1-2 groups	
	Nature	Could model outputs enable action to improve water body health?	Model could tell us how to fix a problem - direct link to enabling action.		Model cannot directly help to enable action to improve water body health.	
	Value	What is the potential value (£) of this theme to the UK water industry? Water UK indicates water companies proposed £105 billion of investment between 2025 and 2030 in draft business plans, therefore 'high' benefit based on c. 5% of this value. Benefits to other stakeholders considered in categories above. Used water company investment as numbers more readily available for assessment, and they are project sponsors.	>£5bn investment over 5 years	£1bn-£5bn investment over 5 years	<£1bn investment over 5 years	
Feasibility	**Previous studies**	Previous study shows AI and ML can answer this question, if not at scale.	Substantial, high quality, localised studies show that this question could be tackled with AI/ML.	>2 previous studies identified	1 or fewer previous studies identified. (Could indicate high innovation potential - not to be discounted, but not all use cases should have no prior work to build on, given the project timescales.)	
	Data availability	Is data to develop models for this theme readily available at scale?	Open, publicly available, national datasets available for long time periods, at regular intervals.	Known private datasets available across UK and Ireland, for long time periods, at regular intervals.	Known private datasets available for small no. of catchments. Data not available at high spatial or temporal resolution.	
			Theme screening – level 2			
Feasibility	**Feasible in RDMAI**	Can models be developed within 18 months, considering the project budget. This will be evaluated through phase 2 of the project for model building blocks.	Yes	Maybe	No	
	Data availability	Is the data density, duration and quality sufficient for model training and testing?	Criteria to be determined by data science team during phase 2.	Criteria to be determined by data science team during phase 2.	Criteria to be determined by data science team during phase 2.	

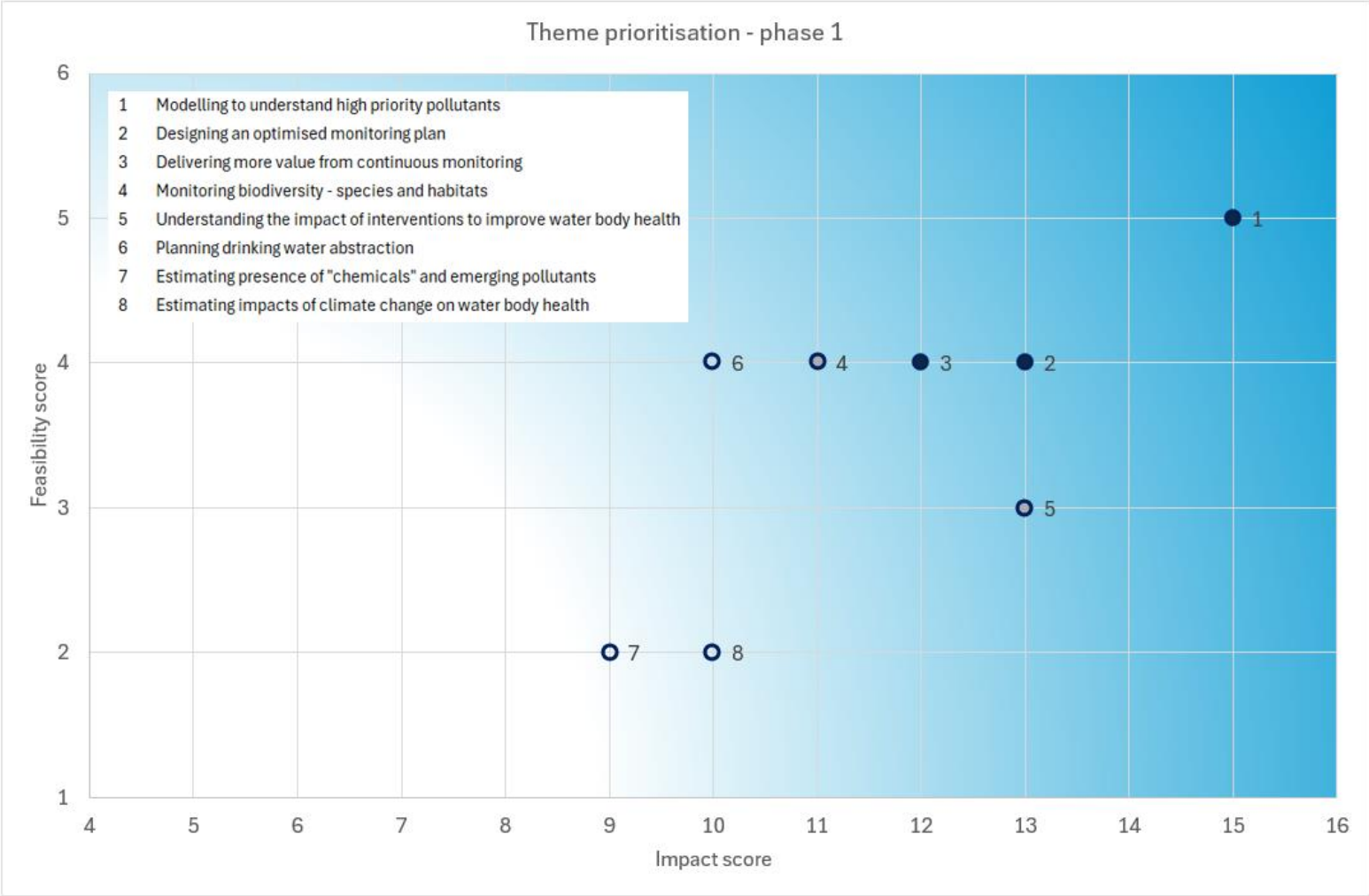

Figure 4: Theme prioritisation results. Themes 1 and 2 fast-tracked to next phase of the project, given high potential impact and feasibility. Theme 3 progresses, with slightly lower priority than themes 1 and 2. Themes 4 and 5 have potentially high impact and/or feasibility but require further investigation to confirm value and feasibility of developing models. Based on resources (staff skills and availability) and project timescales, theme 4 progresses with the lowest priority and will be addressed only if time allows during phase 2 of River Deep Mountain AI.

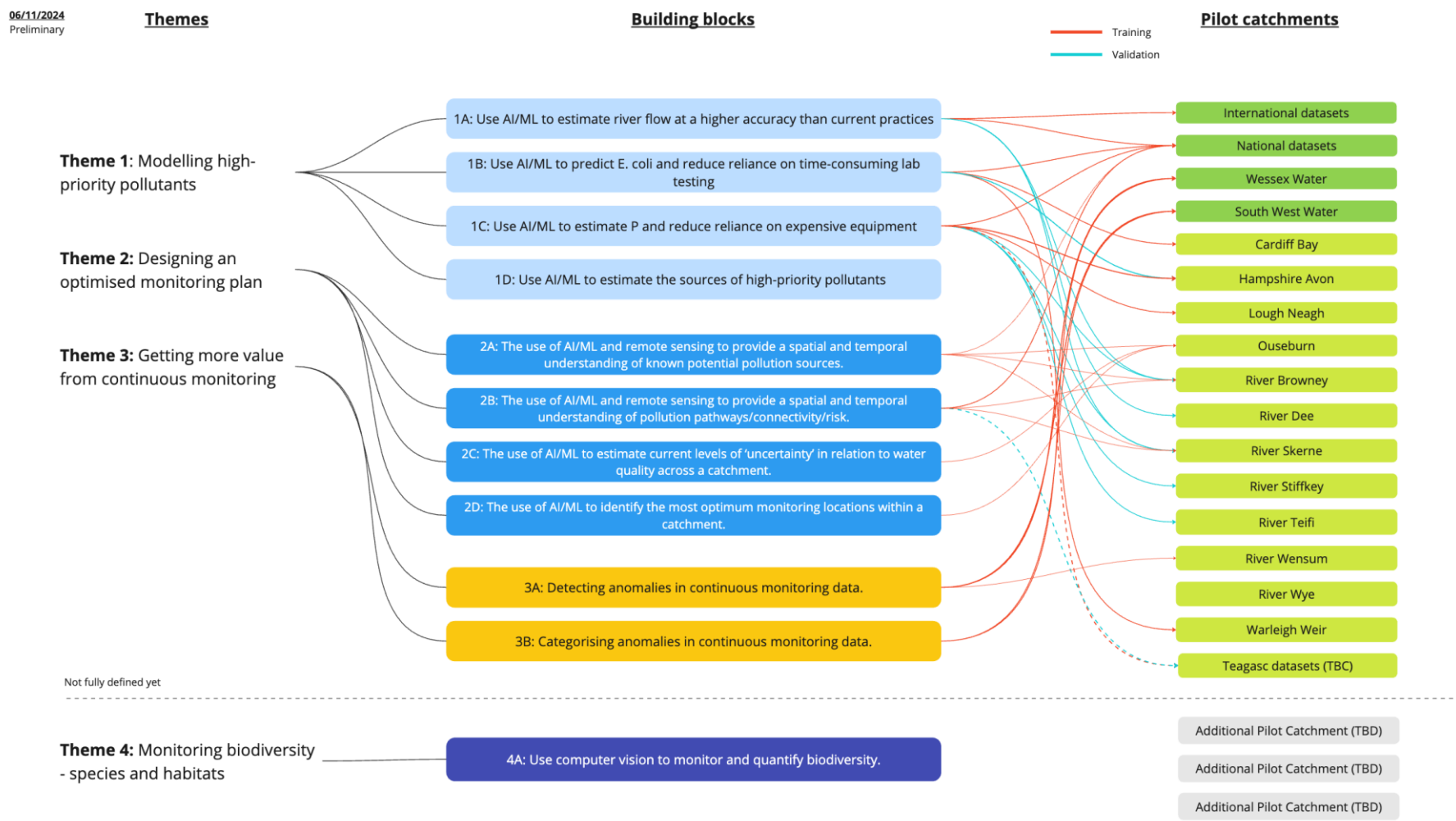

Figure 5: Pilot catchments for each building block – as of October 29th, 2024

Prioritised themes

Following the impact/feasibility screening, the following four themes are developed in more detail below. More detail is provided for the first three themes, as these are progressed with highest priority to the modelling phase of the project.

1. Modelling high-priority pollutants (high priority)
2. Designing an optimised monitoring plan (high priority)
3. Deliver more value from continuous monitoring (medium priority)
4. Monitoring biodiversity – species and habitats (low priority)

Each detailed write-up will consist of three levels (see fig.1):

1. **Challenges and opportunities**: Context, stakeholders, challenges and opportunities.
2. **Building blocks:** Defined problem and value, technical specifications and output.
3. **Use cases**: A description of how the building block(s) can be applied in a real-world setting. These are only descriptions; any development of tools and UI are out of scope.

Pilot catchments

As described in our original submission, an objective of phase 1 of RDMAI has been to find and choose relevant pilot catchments for each building block to support the model development.

To ensure that pilot catchments can support the model development we have considered three relevant aspects:

1. **Data availability:** Relevant water quality and waterbody health data must be available in the catchment in sufficient quantities to train AI/ML models. In some cases, relevant data is available on a regional or national scale, which will increase transferability of models and is therefore preferred.
2. **Access:** As a programme we must have access to relevant stakeholders or partners that can provide local knowledge and access to the pilot catchment. This will ensure that we can access available data and receive feedback and insight if needed. In case we need to collect additional data, having land access is crucial for deploying sensors.
3. **Transferability**: The challenges or water quality data in the catchment must be transferable to other catchments and regions. We want to develop models that can add value across catchments and not address site-specific challenges.

In collaboration with WP1 (project management and governance) and WP3 (catchment and environmental data collection and tech trials) we have outlined a list

of potential pilot catchments and waterbodies for each building block which have the potential to satisfy our needs for developing and testing machine learning models. These catchments are shown in figure 5 (above).

2 Detailed write-ups

In the following, we will go through the individual themes outlined above in detail and describe the challenges and opportunities, building blocks and real-world application(s) for each theme.

Theme 1: Modelling high-priority pollutants (optimise use of existing data)

A key aspect of finding sources of pollution is understanding if, where and when pollution is present in a waterbody. Current approaches to monitoring are expensive and testing for high-priority pollutants such as E. coli and Phosphorous is particularly challenging. An approach to overcome this issue is to use the data that is already being collected in a better way. Applying AI/ML can be a cost-effective and time-efficient way of modelling high-priority pollutants and elevate our monitoring practices.

Overview

Context – Why is this important

Source apportionment modelling, an approach to relate the pollutants with the source, is a key driver of investment decisions in the water sector, with tools such as SAGIS playing a central role in catchment management [1]. These tools are widely used by regulators and water companies to identify pollution sources and assess their contributions to waterbody quality, informing decisions related to environmental regulations, catchment management, and water quality improvement plans. However, current models are often based on limited data (collected bi-weekly, monthly, or annually, and from few locations), leading to inaccurate output, especially for the proportions of diffuse and point-source contributions. Such inaccuracies hinder effective targeting of interventions, resulting in misallocated resources which delays improvements in overall waterbody health.

Lack of trust and inaccurate outputs also complicate fair distribution of responsibilities among stakeholders, making it harder to design equitable and effective mitigation strategies.
These challenges impact long-term water quality management strategies, making it difficult to meet regulatory targets, such as those set under the Water Framework Directive (WFD). Current approaches tend to favour point-source solutions, which

limits the adoption of broader, catchment wide strategies. This, in turn, delays the transition towards sustainable and nature-based solutions (NBS). Furthermore, extensive environmental monitoring, such as via distributed multiparameter sensors, is not a feasible long-term solution due to high costs and logistical challenges, such as frequent calibration requirements and maintenance. This poses significant financial and operational pressure on regulators, water companies and customers.

Incorporating artificial intelligence (AI) and machine learning (ML) technologies can help bridge these gaps by a) minimizing the dependency on extensive monitoring, b) improve the accuracy of existing models, and c) optimising the use of existing data. By enhancing model accuracy and improving the precision of pollution source identification, AI and ML could facilitate evidence driven, targeted water quality interventions, ultimately leading to improved waterbody health.

Stakeholders:

- **Water companies:** Insufficient understanding of pollution sources makes it more challenging for water companies to implement the most efficient/effective mitigation solutions. Consequentially, water companies are enforced to meet tighter discharge limits on phosphorous, rather than enabled to pursue interventions on a catchment scale.
- **Regulators**: Resources are limited, and monitoring is expensive and time consuming. Inaccurate source apportionment modelling hinders fair and effective enforcement of water quality standards across sectors and makes planning and evaluation of environmental policies challenging.
- **Local communities**: Safe and healthy waterbodies are essential for public health, recreation and biodiversity. Unnecessary spending on ineffective or misdirected interventions intensifies the financial burden on customers.

Challenges:

- The current lack of water quality and flow data is the reason why the data-reliant SAGIS model output can be inaccurate. Low number of gauging stations are used to infer flow data across large areas. As an example, only 9 gauges are used to run SAGIS across more than 500km of rivers in the North-East (C5, notes). Filling data gaps is expensive. Easily £300,000 to monitor a single parameter in a waterbody (C5, W28, Interviews).
- *Escherichia coli (E. coli)* and Phosphorous are high priority for regulators - monitoring P and *E. coli.* is time consuming and expensive. With continuous phosphate sensors costing up to £50,000 per sensor (NWL).
- Water companies make significant investments in extensive data collection schemes to challenge model output.

"Flow is underrepresented in the Skerne, leading to glaring gaps in SAGIS outputs. Typically, the model will use surrogate river flow data, which is only sense checked every 5 years." (C5, interviews, notes)

"[…] we actually did a huge catchment monitoring program to improve the quality of the models because the models were showing a really significant impact of the wastewater treatment works and the catchment partnership just said no chance. It's not really them, it's agriculture that's causing the problem here […] we spent about £350,000 and did quite a lot of detailed sampling right down the river to improve the quality of that model." (W28, Interviews)

"[It] takes c. 1 week to get sample results back from labs, by which time it's too late to work out where pollution came from." (W2, Interviews, paraphrasing)

Opportunities

Our main hypothesis for this theme is that by optimising how we model and use the data we already have; we can improve model accuracy and reliability for increased understanding of the pollutants' sources and impact.

We have defined four opportunities where AI/ML can be applied to overcome some of the specific challenges related to this theme. A key element that permeates through all is the use of spatial and temporal data together to give a more holistic understanding of pollution, sources and impact.

"Surrogate for phosphate – would be a gamechanger" (C5, interviews, notes)

"[How could AI help] link bacteria concentrations to other parameters so that it's possible to measure parameters like E. coli in 'real time'." (W13, Interviews, paraphrasing)

We want to address the lack of flow data that is impacting the output of the SAGIS model, as well as the challenges tied to time-consuming and expensive monitoring of *E. coli.* and phosphorous that often delays or impedes decision-making related to those pollutants. Lastly, we want to investigate how well AI/ML can be used to model the travel-time and dilution of pollutants in waterbodies.

This has led to the definition of four potential AI/ML building blocks:

- **Building block 1A:** Use AI/ML to estimate river flow data at a higher accuracy than current surrogate practices.
- **Building block 1B:** Use AI/ML to predict *E. coli.* to reduce reliance on time-consuming lab testing.

- **Building block 1C:** Use AI/ML to estimate phosphorous to reduce reliance on expensive equipment.
- **Building block 1D:** Use AI/ML to estimate the sources of high-priority pollutants.

The problems, values and objectives of the following building blocks are based on stakeholder research carried out in phase 1. The technical specifications (including data and pilot catchments) will be further developed in collaboration with subject-matter experts and data scientists and are subject to change during phase 2 of the project.

Although interesting, cyanobacteria was ultimately excluded, due to inherent re-active aspects of addressing harmful algae blooms (HABs). A great piece of work was done on this in the early stages of the project and findings are detailed in a summary report (see appendix).

Building blocks

Building block 1A: Use AI/ML to estimate river flow data at a higher accuracy than current surrogate practices.

Defined problem and value

- Flow data is limited in most catchments. Flow is a key parameter going into the SAGIS source apportionment model, and essential when trying to understand where pollution is coming from and pollutant loads.
- Currently data gaps are filled using surrogate flow data from other similar rivers. These surrogates are sense-checked every 5 years. This approach comes with its own disadvantages.
- To increase the accuracy of SAGIS in individual waterbodies, stakeholders undertake expensive 12-month monitoring programmes of flow data. Monitoring campaigns like this can easily cost upwards of £300,000.
- An AI/ML model that can estimate flow data at a higher accuracy than current surrogate practices will improve current modelling and enable water companies and regulators to get a better, faster and cheaper understanding of where pollution is coming from.

Technical specifications (preliminary)

- **Modelling techniques**: AI/ML model that considers several layers of data in a catchment to estimate flow based on historical data. Use data that indicates changes in water flow of a river (rain, CSO, effluent), weather and static catchment attributes (e.g. high-altitude upland, lowland, soil type, vegetations,

etc.) together with real-time flow data to train a model to estimate flow.

- **Suggested data inputs**:

  Dynamic inputs:
  - Near real-time river flow data: Hydrology Data Explorer, Estream, CAMELS, HYSETS
  - Rainfall and meteorology: MET, Hydrology Data Explorer, CAMELS, HYSETS
  - CSO/Effluent: Water Companies

  Static catchment attributes
  - River width and depth: CEH
  - Slope and topography: EA Lidar data, CAMELS, HYSETS
  - Land cover: CROME, Rural Payment Agency, CEH, Estream, CAMELS, Living England Habitat Map
  - Vegetation: Estream
  - Soil and geology: Estream

- **Data sources**: Hydrology Data Explorer, MET data, Water Companies, STREAM, Crop Map of England (CROME), Rural Payment Agency, EStreams, CEH, CAMELS, HYSETS, Soil data.

- **(Pilot) Catchments**: Phase 2: Trained on as many catchments as possible in the Hydrology Data Explorer where river flow and rainfall data are available real-time for the past 20 years. Phase 3: Validated and improved in a pilot catchment where new flow data is currently being collected (e.g. Skerne, Browney or Dee).

- **Reference(s)**:
  - **Google Research**: Has done something similar to predict extreme floods in ungauged rivers. The model they have used to build their flow model is open source and available via Kratzert et al (2019).
  - **Andrew Johnson**, Data Scientist, The Rivers Trust: Has worked on modelling the flow of the river Thames and connected basins. Preliminary work, but relevant to include in this work.
  - **Lily Gouldsbrough**, Data Scientist, CEH: Working with flow modelling.

Output (preliminary)

Model outputs will be defined more clearly by theme leads during phase 2. Below is suggested preliminary outputs:

- **Output format**: Resembling data output coming from an EA controlled flow monitor / gauge.
- **Key parameters**: Estimated flow in m^3/s, time, date, location.
- **Model evaluation**: Phase 2: Validate against flow collected in catchments outside of the training data. Phase 3: Validate against current surrogate modelling (if possible).

Use cases:
The real-world application for stakeholders is for the models within this building block:

- **Water companies and charities:** Having an AI-driven flow estimation model will strengthen water companies and charities ability to estimate river flow in ungauged rivers so that they can fill in current data gaps in areas where monitoring missing or limited. This will help initiatives such as Thriving Catchments and others to get a better understanding of how river flow fluctuates, enabling them to more accurately estimate where pollution is coming from without increasing cost of monitoring. Filling these data gaps will also enable water companies, and partners, to strengthen the outputs of the SAGIS model in England to get a better understanding of source apportionment. Alternatively, estimating the river flow in several catchments and testing the effects that additional flow data would have on SAGIS output could also be used to prioritise which catchments to run an intensive flow monitoring programme reducing the cost and uncertainty associated with monitoring.
- **Regulators**: Having an AI-driven flow estimation model will enable regulators to increase their coverage of river flow monitoring beyond gauged rivers. This can be used to fill in data gaps on publicly available datasets, improve understanding of source apportionment and improve flood forecasting tools.

Building block 1B: Use AI/ML to predict *E. coli.* to reduce reliance on time-consuming lab testing

Defined problem and value

- Monitoring *E. coli.* is currently constrained by the time it takes to complete a lab test and identify *E. coli.* colony forming units per 100ml (CFU/100ml). This is a process that, depending on logistics, takes at least 48 hours but also up to a full week.
- *E. Coli.* is a high-priority pollutant for regulators to monitor. It plays a key role in classifying and regulating designated bathing waters.
- In most cases, knowing that bathing water contained *E. coli.* 48 hours ago is not very useful to act or warn.

- An AI/ML model that can use parameters more easily accessible and measurable to predict the concentration of *E. coli.* in bathing waters will enable water companies and regulators to make better decisions, reduce costs on monitoring, and inform local communities on the quality of bathing waters.

Technical specifications (preliminary)

- **Modelling techniques:** AI/ML model that considers several layers of data in a catchment to estimate (and predict) *E. coli.* (CFU).

- **Suggested data inputs:** Water sources, weather, physio-chemical water data, static catchment attributes.
    - Lab samples of *E. coli.* Levels – if possible, also sensor data (Proteus sonde).
    - Water sources: Rainfall, CSO/STW.
    - Weather: Temperature, humidity, more.
    - Variables collected through section 82: pH, temperature, ammonia, dissolved oxygen, turbidity
    - Additional physio-chemical water data: Conductivity, total suspended solids, BOD, COD, etc
    - Coastal data: salinity, tidal, etc.
    - River data: Flow

  Static catchment attributes
    - River width and depth: CEH
    - Slope and topography: EA Lidar data, CAMELS, HYSETS
    - Land cover: CROME, Rural Payment Agency, CEH, Estream, CAMELS, Living England Habitat Map
    - Vegetation: Estream
    - Soil and geology: Estream

- **Data sources:**
    - Hydrology Data Explorer, MET data, Swimfo/EA, Water Companies, STREAM, Crop Map of England (CROME), Rural Payment Agency, CEH.

- **(Pilot) catchments:**
    - **Data availability:** E. coli (CFU/100ml) data is limited on a national scale making it difficult to develop transferable models. A limited set of individual catchments have data available for site-specific model development. Static catchment attributes are not readily available and might require labelling and pre-processing.
    - **Estuarine bathing waters:** Alternatively, explore if EA bathing water monitoring data across English estuarine bathing waters can be used to

develop a transferable model. Data here is limited to 20 samples a year per location.
    - **Inland bathing waters:** Wessex Water (Warleigh Weir) has 2+ years of bacteria data and sonde data available. This would most likely result in a site-specific model.

- **Reference(s):** Wessex Water and UnifAI has completed trials in Warleigh Weir using a similar approach as above, but without static catchment attributes.

Output (preliminary)
Model outputs will be defined more clearly by theme leads during phase 2. Below is suggested preliminary outputs:

- **Output format**: With key parameters such as predicted *E. coli.* (cfu), time, date, location.
- **Model evaluation**: Phase 2: Validate against available *E. coli.* data.

Use cases:
The real-world application for stakeholders is for the models within this building block:

- **Regulators**: Having an AI-driven E. Coli prediction model, will enable regulators to strengthen current pollution risk forecasts for bathing waters (such as Swimfo) so that they can provide frequent and reliable forecasts across all bathing waters.
- **Water companies**: Having an AI-driven E. Coli prediction model, will enable water companies to bypass time-consuming lab samples and provide frequent forecasts of pollution risks at designated and undesignated bathing sites in their local area. Ultimately, a better understanding of E. Coli contamination can also provide insight into potential sources of contamination and a better understanding of river health overtime.
- **Communities**: An AI-driven E. Coli predictor will enable local communities, such as open water swimmers, rowing clubs, and others, to receive E. Coli pollution forecasts that will support them in making informed decisions on a day-to-day basis whether to go in the water. From a communicative point of view a forecast of E. Coli can also be used to communicate the health of rivers to local communities.

Building block 1C: Use AI/ML to estimate phosphate to reduce reliance on expensive equipment.

Defined problem and value

- Monitoring phosphorous currently requires lab testing and wet chemistry and continuous monitoring is only available using expensive lab-based analysers.

- Phosphorous is a high-priority pollutant for regulators and water companies to monitor. Phosphorous levels play a key role in the ecological classification of a waterbody.
- An AI/ML model that can use more easily accessible and measurable proxy parameters to estimate the concentration of phosphorous in waterbodies will enable water companies and regulators to get better insight into the impact of phosphorous at a lower cost and higher frequency.

Technical specifications (preliminary)

- **Modelling techniques**: AI/ML model that considers several layers of data in a catchment to estimate phosphorous using proxy parameters.

- **Suggested data inputs:** Focus should be on parameters more common and easily monitorable than phosphorous.

  Dynamic variables:
  - Variables collected through section 82: pH, temperature, ammonia, dissolved oxygen, turbidity.
  - Rainfall and meterology: MET, Hydrology Data Explorer, CAMELS, HYSETS,
  - Additional physio-chemical: conductivity, total suspended solids, COD, BOD, etc.
  - Flow data.

  Static catchment attributes
  - River width and depth: CEH
  - Slope and topography: EA Lidar data, CAMELS, HYSETS
  - Land cover: CROME, Rural Payment Agency, CEH, Estream, CAMELS, Living England Habitat Map
  - Vegetation: Estream
  - Soil and geology: Estream

- **Data sources**
  - EA, Water companies, STREAM.

- **(Pilot) catchments:**
  - **Data availability**: High-frequency P data (mg/l) is limited on a national scale making it difficult to develop transferable models. A limited set of individual catchments have data available for site-specific model development. Static catchment attributes are not readily available and might require labelling and pre-processing.
  - **National dataset:** Phosphorous data is available in low-frequency for England (Defra/WFD).

- **Catchment datasets:** Phosphorous data will potentially be available in higher frequency from Lough Neagh (NI Water), Hampshire Avon (Southern Water), River Browney (NWL), River Skerne (NWL), River Stiffkey (Anglian Water), Teagasc (Northern Ireland)

Output (preliminary)
Model outputs will be defined more clearly by theme leads during phase 2. Below is suggested preliminary outputs:

- **Output format**: With key parameters such as estimated phosphorous, time, date, location.

Use cases
The real-world application for stakeholders is for the models within this building block will enable:

- **Regulators** and **water companies** to predict phosphates in waterbodies using proxy parameters and thereby bypass expensive monitoring equipment and extend their phosphate monitoring to cover larger catchment areas. This will make it easier, faster and cheaper to locate areas where expensive monitoring could be deployed for investigation purposes and ultimately help spot trends and hotspots of phosphate pollution in a catchment.

Building block 1D: Use AI/ML to estimate the sources of high-priority pollutants

Defined problem and value

- Source detection is challenging due to lack of data points and the complexity of waterbodies leading to ineffective interventions.
- Modelling pollutant travel time is a key component for identifying pollutant sources in rivers because it enables the tracing of contaminants from their origin to affected areas.
- The movement of pollutants is influenced by factors like flow rates, UV-exposure, temperature and pH, and other pollutant characteristics which affects their decay and transport.
- Using AI/ML to model pollutant travel time can improve source detection by improving the ability to differentiate between point and diffuse sources
- Our hypothesis is that combining environmental factors with travel dynamics can improve our ability to pinpoint pollution hotspots.

Technical specifications (preliminary)

- **Modelling techniques:** AI/ML model that considers several layers of data in a catchment to model pollutant travel time.

- **Suggested data inputs:** environmental data including rainfall, temperature, flow rates, conductivity and pH.

- **References**: Alison Furber from the Rivers Trust and Andy Young from Dundee University has done work on this for CSO/STW.

Output (preliminary)

- **Output format**: GIS layer(s) visualising pollution dynamics within a catchment - spatially and temporally.
- **Model evaluation**: To be detailed in phase 2.

Use cases:

The real-world application for stakeholders is for the models within this building block to enable:

- **Water companies** to trace pollutants back through the waterbody to strengthen their understanding of pollution sources and hotspots in a catchment. This can support catchment-based interventions and efforts to reduce pollution, from both point-source and non-point source pollution sources and help widen the scope of investigations into pollution sources and source apportionment. An AI-driven model can be a supplementary model to use together with SAGIS to enhance catchment-wide interventions.
- Models that support the efforts of determining sources of pollution will also be useful and beneficial for stakeholders across the catchment to find common ground and support decision making on suitable interventions and where to intervene.

Theme 2: Designing an optimised monitoring plan

To make informed decisions when designing and implementing water quality interventions, stakeholders must understand the current state of water body health within a catchment.

In many catchments, existing datasets often lack the accuracy and reliability to adequately inform such decision processes. Consequently, stakeholders must often implement additional water quality monitoring schemes, before interventions can be designed and constructed.

Due to the heterogeneity of catchments and the variety of water quality issues that can be affecting them, understanding where and what to monitor within a catchment can be challenging and expensive to work out.

Therefore, developing an AI/ML model that can assist stakeholders in the design and implementation of water quality monitoring plans has the potential to save significant financial costs and help improve the sector's overall understanding of catchment waterbody health.

Overview

Context – Why is it important

For stakeholders to decide what and where interventions are needed to improve waterbody health, an understanding of the current catchment 'situation' must be known. Whilst many catchments have some level of existing water quality datasets, it can be challenging to understand the accuracy and reliability of existing water quality datasets within a catchment. Consequently, before a water quality intervention can be implemented, most catchment stakeholders must undertake preliminary water quality monitoring programmes, so that the decision process is informed by accurate and reliable data.

Designing and implementing preliminary water quality monitoring programmes is often time consuming and expensive. In many cases, multiple in-person site surveys are required during the initial stages of a project, to help inform where water quality sampling or monitoring 'schemes' should be implemented within a catchment. A single in-person basic site walk-over will generally take *"about two days to do 500 metres of river"* and can cost "*upwards of £200 per day*" (C2, Interviews). As such, projects within larger catchment areas will incur significant financial costs before any actual water quality monitoring has even begun.

Furthermore, continuous water quality monitoring is costly, limiting the number of monitors that can be deployed across a catchment. Given that water quality variables can fluctuate significantly over short timescales and across different spatial locations, monitoring schemes often lack the precision needed to capture these changes accurately. While continuous monitoring captures changes over time, it often misses spatial variability across different river sections, especially those distant from monitoring stations. Drones, conversely, can survey spatial changes effectively, but lack the temporal continuity needed for tracking water quality dynamics over time. To generate accurate and comprehensive water quality insights, stakeholders are depending on cost-effective, optimised monitoring schemes that capture the complex dynamics of entire catchments – addressing challenges of limited spatial and temporal sampling.

Following the recent regulatory changes associated with Section 82 of the Environment Act (2021), English water companies will be investing in significant continuous water quality monitoring of five water quality parameters upstream and downstream of storm overflow and wastewater treatment works effluent. This theme

will investigate whether additional value can be realised by optimising the placement of these monitors and by using the continuous data to inform other monitoring, such as sampling campaigns and drone surveys.

Although section 82 of the Environment Act (2021) only applies to England, we find it fair to assume that water companies and other stakeholders, outside of England, will be increasing their continuous monitoring of water quality in in the future. Thus, making this theme relevant outside of England as well.

Stakeholders

- **Water Companies:** to enable quick, reliable and efficient design of water quality monitoring plans, to assist in the implementation of water quality interventions. Relevant for catchment-based approaches and catchment nutrient balancing efforts across water company value-chain; wastewater and clean water.
- **Charities:** help build a clear understanding of existing data reliability/accuracy along a river water body, to help target areas that could benefit from citizen science projects or additional water quality monitoring schemes.
- **Regulators:** help build a clear understanding of existing data reliability/accuracy along a river waterbody, to help identify whether additional monitoring schemes may be needed to meet regulatory requirements.

Challenges

The primary challenge this theme aims to address is the significant financial costs associated with the design and implementation of a water quality monitoring scheme by stakeholders. These financial costs can be loosely separated into two themes:

(1) The cost of preliminary site surveys and the design of the monitoring scheme.
(2) The capital and operational cost of continuous monitoring equipment or ongoing sampling activities.

As the capital and operational costs associated with in-situ water quality monitors can be high, most catchment monitoring schemes rely on a relatively small number of monitoring sites. Consequently, stakeholders can also face challenges when attempting to understand water quality 'in between' monitoring sites. In other words, the interpolation of water quality data is often inaccurate and unreliable, making it difficult to understand how water quality varies within a waterbody, unless a significant number of monitors are installed. Furthermore, if monitors are in the 'wrong' location, the usability and quality of data they produce can be severely limited, minimising their cost-effectiveness.

Opportunities
The main aim of this theme is to use AI/ML and remote sensing to help reduce the financial costs associated with designing and implementing a water quality monitoring scheme within a catchment. The outputs from this theme should help relevant stakeholders understand the most cost-effective way of monitoring the target catchment, by highlighting the most efficient way a catchment can be monitored using a limited number of monitors. For example, the AI/ML model could inform what level of understanding could be achieved, if 'x' number of monitors were installed. In addition, the AI/ML model should help stakeholders understand where to re-locate existing monitoring sites to improve overall accuracy and understanding and catchment water quality. Similarly, the value associated with the implementation of additional sensors will be quantified.

"When we are looking at river catchments, understanding where we can get the most value and make the biggest difference, spatially this is where we can make the biggest difference and improvement to water quality [...]." (C7, 41:53, Interviews)

This theme also provides an opportunity to quantify and compare the value of continuous monitoring points versus sampling programmes. Whilst sampling schemes provide an adequate understanding of gradual changes over time, continuous monitoring excels in providing data during 'irregular' events. Being able to quantify the optimum use for each approach will provide valuable insight when planning catchment monitoring schemes.

This has led to the definition of four potential AI/ML building blocks, that could be used to accomplish the overall aim:

- **Building block 2A:** Use AI/ML and remote sensing to provide a spatial and temporal understanding of potential pollution sources.
- **Building block 2B:** Use AI/ML and remote sensing to provide a spatial and temporal understanding of pollution pathways/connectivity/risk.
- **Building block 2C:** Use AI/ML to estimate current levels of 'uncertainty' in relation to water quality across a catchment.
- **Building block 2D:** Use AI/ML to identify the most optimum monitoring locations within a catchment.

The problems, values and objectives of the following building blocks are based on stakeholder research carried out in phase 1. The technical specifications (including data and pilot catchments) will be further developed in collaboration with subject-matter experts and data scientists and are subject to change during phase 2 of the project.

Building blocks

Building block 2A – Identify and Map Potential Sources of Pollution

Defined problem and value:

Understanding potential sources of pollution in a catchment currently requires time-consuming on-the-ground walkovers or extensive work with GIS data. The aim of this building block is to use AI/ML and remote sensing to identify and map pollution sources in a time efficient way.

If possible, known sources of pollution should be distinguished from potential sources of pollution. Irregular pollution and permitted pollution sources should be differentiated clearly. Similarly, the distinction between point source and diffuse source pollution should be highlighted.

Sources of pollution could include storm overflows, wastewater treatment work effluents, highway runoff, agricultural runoff (livestock, slurry tanks, septic tanks), industrial discharges and camp sites.

Technical specifications:

- **Modelling techniques:** Remote sensing and computer vision (AI/ML) that automates the process of finding potential sources of pollution. For computer vision models, 'papers with code' may be a helpful source of information.
- **Potential data inputs:** satellite imagery, land cover datasets, water company asset data, agricultural asset data, crop cover maps, BGS datasets, EA discharge permits.
    - Limitations of remote monitoring data (e.g. cloud cover, frequency of data and width of water body, etc.) should be considered.
    - Possible image labelling team needed.

Potential Pilot Catchments:

- River Ouseburn, River Browney, River Skerne.

Reference(s):

- Pollutant Impact Potential (PIP) maps developed by EPA in Ireland: https://www.catchments.ie/next-generation-pollution-impact-potential-maps-launched/
- ALERT Tool developed by Environment Agency.

Output:

Model outputs will be defined more clearly by theme leads during phase 2. Below is suggested preliminary outputs:

- **Output Format:** The primary output from this building block will be a GIS layer containing labelled pollution sources.
    - The building block could be used to help develop a standard approach to desktop surveys.
    - Important to categorise potential sources of pollution.

Use cases:
The real-world application for stakeholders is for the models within this building block to enable:

- **Regulators:** Having a model to cheaply and efficiently identify potential sources of pollution in a catchment will allow regulators to gauge what the key risks to waterbody health are within a catchment, without the need for extensive in-person surveys. In addition, the model outputs may help identify pollution sources that have not previously been accounted for. Utilising this model as a standard approach for desktop surveys would allow for more comparability between different catchments, helping to highlight key areas of risk.
- **Water companies and charities**: Having a model that identifies potential pollution risks within a catchment will enable water companies and charities to save time and money associated with extensive in-person catchment surveys. The model outputs will give stakeholders an understanding of potential 'high-risk' zones and highlight what issues need to be addressed with water quality interventions.

Building block 2B – Identify and Map Possible Pollution Pathways/Connectivity

Defined problem and value:
Understanding pollution pathways and connectivity in a catchment currently requires on-the-ground walkovers during rain events and/or analysis using GIS software.

The aim of this building block is to use AI/ML and remote sensing to identify high risk areas with high connectivity and pollution pathways:

- To understand the impact or risk each potential pollution sources poses to waterbody health, pollution pathways must be identified and visualised both spatially and temporally.
- To visualise potential 'pollution hotspots' within a catchment area and help identify the most at-risk parts of a waterbody.
- Potential pollution pathways could include runoff, river networks, sub-surface pathways, etc.

- The model outputs should be able to help visualise how pollution pathways have changed over time (e.g. via land use change). This theme also has the potential to add value by providing a reduction in the uncertainty associated with water quality estimations.

Technical specifications:

- **Modelling techniques:** Remote sensing and computer vision (AI/ML). May already be mature/existing products available to use.

  This building block will predominantly rely on 'non-water quality specific' datasets, such as soil type, slope, meteorological datasets, etc…

- **Potential data inputs:** satellite imagery, land cover datasets, soil type, slope, meteorological datasets, etc.
  - May be (at least partly) dependent on building block 2A.
  - LIDAR (or level data) could prove useful.

Potential Pilot Catchments:

- Possibly in Ireland (Daniel Hawtree point of contact) 11 years WQ data (P/N ++) in six agricultural sites.

Reference(s):

- Pollutant Impact Potential (PIP) maps developed by EPA in Ireland: https://www.catchments.ie/next-generation-pollution-impact-potential-maps-launched/
- ALERT Tool developed by Environment Agency.

Output:

Model outputs will be defined more clearly by theme leads during phase 2. Below is suggested preliminary outputs:

- **Output Format:** The primary output from this building block should be GIS layers visualising the pathways through which pollution sources are entering the river waterbody. Ideally, each pollution pathway should also have an associated level of 'risk' (i.e. how much of a risk does this pollution pathway pose to waterbody health).
- **Model output:** Try and turn all our data into a connectivity graph (in the mathematical sense) with weights for each parameter and lengths, locations etc

- **Model Evaluation Criteria:**

- Validate the output of the model with traditional methods of doing this: on-the-ground walkovers during rain events (assume this method is the ground-truth method).
- Compare time to use the AI/ML approach vs doing the walkover.
- Quantify cost saving as per time saving above.
- Compare output of the AI/ML model to output using GIS software approach.

Use cases:
The real-world application for stakeholders is for the models within this building block to enable:

- **Regulators, water companies and charities**: Similarly to building block 2A, model outputs detailing how and where pollution is entering catchment waterbodies will give regulators, water companies and charities a better understanding of what pollutants are entering the catchment river system in a particular location. Additionally, the associated level of 'risk' assigned to each pollution pathway will enable stakeholders to prioritise interventions on the worst affected areas. Like building block 2C, utilising this model as a standard approach would allow for more comparability between different catchments, helping to highlight key areas of risk.

Building block 2C – Estimate current levels of 'accuracy/reliability/uncertainty' of water quality across a catchment

Defined problem and value:

In order to produce a set of 'optimum monitoring locations' that can be utilised by the model user, the level of uncertainty associated with current water quality estimations across the catchment must be understood. Once this level of uncertainty can be modelled for a catchment, an optimisation process can then be undertaken to reduce the uncertainty as much as possible. Uncertainty may need to be associated with each individual WQ parameters. Although perhaps a total catchment uncertainty value could also be calculated in some manner?

Technical specifications:

- **Potential modelling techniques:**
  - Interpolate river parameters along the river network, calculating associated uncertainty as a function of the position.
  - The uncertainty along the river network can then be 'summed'/'interpolated' to generate a total catchment uncertainty

value. This could be weighted by importance / risk associated with particular locations.
    - Important to account for seasonal variation.
- **Potential models:**
    - Bayesian Mixed Effects Model – Interpolating and extrapolating temperature. Data needed for this approach. (Andrew Johnson to provide detail on paper and approach)
    - Physics restricted simulations - e.g. sDNA OCN simulation (Andrew Johnson to provide details on paper and approach).

Potential Pilot Catchments:

- Need a selection of 'data rich' and 'data sparse' catchments. Perhaps a 3:1 ratio respectively.
- Suggested pilot catchments: River Ouseburn, River Browney, River Skerne.

Output:
Model outputs will be defined more clearly by theme leads during phase 2. Below is suggested preliminary outputs:

- **Output Format:** A spatial and temporal representation of water quality across the catchment should be produced. A similar spatial and temporal representation of 'uncertainty' associated with this estimation should be produced.

Use cases:
The real-world application for stakeholders is for the models within this building block to enable:

- **Regulators:** having a 'water quality uncertainty' model will give regulators a high-level, spatial understanding of the 'gaps' in existing water quality monitoring programmes. In other words, the model outputs will help show regulators which areas of a catchment river system are lacking water quality information/data. This will help regulators decide whether further water quality monitoring may be needed.
- **Water companies and charities:** having a 'water quality uncertainty' spatial representation of a river within a catchment, will allow water companies and charities to understand where current water quality monitoring schemes may be lacking. This will allow water companies and charities to focus any additional future monitoring programmes (sensors, sampling or citizen science) in these areas of 'high uncertainty', helping to improve the overall, wholistic understanding of water quality within a catchment.

Building block 2D - Identify the most optimum monitoring locations within a catchment

Defined problem and value:

Stakeholders face significant time and financial outgoings when attempting to design and implement a water quality monitoring scheme within a catchment. To reduce these initial design costs, this building block should produce an AI/ML model that can recommend the optimum number of sensors/monitors needed within the catchment, to produce a water quality dataset with particular level of 'accuracy/reliability/uncertainty'. In other words, how many data points are needed to accurately understand the water quality within a catchment. The model should recommend the optimum location for the monitoring points. Additionally, the model should be able to optimise existing monitoring schemes, to help improve overall 'accuracy' in the catchment-wide water quality dataset.

Technical specifications:

- **Potential modelling techniques:**
  - Optimally, this building block can utilise the outputs from previous building blocks within this theme.
  - The sum / 'integrated uncertainty' from building block 2C is calculated for each hypothetical monitoring scheme. Uncertainty should then be minimised (gradient descent).
  - The user should be able to specify how many monitors they want to install and/or the level of 'accuracy/reliability/uncertainty' they wish to produce with their proposed monitoring scheme.
  - Possible modelling process: The sum / 'integrated uncertainty' from building block 2C is calculated for each hypothetical monitoring scheme. Uncertainty should then be minimised (gradient descent).

- **Potential data inputs:** satellite imagery, DEM, land cover data, land accessibility data, river morphology, terrain, existing monitoring schemes, tree cover, depths, etc. Local, site-specific factors may be needed.
  - Will require additional data collection from Xylem: Over-monitoring in order to test the model. I.e. high density of monitoring that would normally be affordable. See how much the model deteriorates as data is removed.
  - Ideally, real world data from water companies should be used to train and/or validate the model, otherwise stakeholders may be reluctant to trust the outputs.

Potential Pilot Catchments:

- Suggested pilot catchments: River Ouseburn, River Browney, River Skerne.

Output:
Model outputs will be defined more clearly by theme leads during phase 2. Below is suggested preliminary outputs:

- **Output Format:** The output produced from this building block should be a clear representation of the most optimum way (cost and coverage) to monitor a given catchment for a given pollutant (to be decided in phase 2). This will likely include a spatial representation of where monitors should be located. In other words, the user should be able to clearly understand how the implementation of their proposed monitoring scheme would impact the 'reliability/accuracy/uncertainty' of water quality estimations across the catchment area, thus informing the user whether any further monitoring may be required to reach a satisfactory level of reliability.

Use cases:
The real-world application for stakeholders is for the models within this building block to enable:

- **Regulators, water companies and charities:** having a clear understanding of the most 'optimum way' to monitor a catchment will help ensure that regulators, water companies and charities understand the most cost-effective way to monitor water quality in a given catchment, to improve water quality understanding as much as possible. This will provide the information needed to understand if and/or how existing monitoring sites/programmes should be relocated to improve holistic catchment understanding. Furthermore, outputs from this model will provide stakeholders with details on where any additional monitoring should be placed to maximise benefits. Conversely, the model outputs should also be able to highlight the determinantal effect of removing/stopping a particular water quality monitoring programme. Thus, outputs from building block 2D will enable stakeholders to understand the relationship between number of monitoring sites/programmes and water quality understanding/uncertainty. This information can be used to help inform whether further monitoring is needed within a catchment or whether funding may be better spent elsewhere.

Following on from this project, outputs from 2D could be used in a tool that generates an interactive map which initially suggests monitoring and sampling locations. Then if the user were to move a monitoring location, the level of uncertainty within the surrounding river reaches would change as a reflection of that. Please note that the models developed within the RDMAI project are intended to serve as the core engines or components powering such tools and use-cases. The

scope of RDMAI does not extend to the development of user interfaces or interactive elements.

Theme 3: Getting more value out of continuous monitoring

The overall objective of this theme is to use AI/ML models to increase the value and accuracy of the continuous water quality data coming out from section 82 of the Environment Act (2021).

We believe that AI/ML can be a way of making sure that water companies and regulators get the most value out of this increased monitoring.

The output from this theme should improve stakeholders' ability to sense-check and understand section 82 data in a time-efficient and thorough manner.

Overview

Context – Why is this important

To better understand the impact of discharges from sewage treatment works and storm overflows on waterbodies, the Department of Environment, Food and Rural Affairs has put a statutory requirement on English water companies to continuously monitoring upstream and downstream of their assets (Environment Act 2021, Section 82).

This will increase current OPEX/CAPEX related to monitoring and create unprecedented levels of incoming real-time data for both water companies and regulators.

This theme is initially most relevant for water companies and regulators in England, but we assume that moving forward more stakeholders will increase their capabilities for continuous monitoring of water quality, making it important for all to sense-check vast amounts of data.

Stakeholders

- **Water companies**: Defra's provisional guidance on implementing section 82 of the Environment Act (2021) requires water companies in England to publish continuous monitoring data within 1 hour. This puts additional pressure on water companies to sense-check data.
- **Regulators**: Like water companies, regulators are expected to publish and use continuous water quality data coming in from water companies.

Challenges

- Increased capital cost of acquiring and deploying sensors, as well as setting up live dashboards, visualisations etc. to adhere to statutory requirements.
- Increased operational cost to maintain sensors.
- Unprecedented amounts of incoming data that must be sense-checked and published within 1 hour every day.
- Together these put water companies and regulators under pressure to handle sensors, data and sense-checking.

Opportunities

Our main hypothesis for this theme is that by utilising AI and ML we can support water companies in adhering to the provisional guidance on implementing section 82 of the Environment Act (Defra, 2023) where water companies are required to publish continuous water quality data within 1 hour.

"Let's just talk about the data for a minute. We have to publish these results within one hour. On a customer facing dashboard [...] To do that, we need to be confident that the results are accurate, that the results, there's no false flags in there. We need to screen it." (W3, Interviews)

[...] We can do it with humans or current technology, but definitely the world of AI will play a massive part in achieving that one-hour publication of that data with confidence." (W3, Interviews)

We have defined two specific opportunities where AI/ML can be applied to overcome some of the specific challenges related to this theme.

- **Building block 3A:** Detecting anomalies in continuous monitoring data.
- **Building block 3B:** Categorising anomalies in continuous monitoring data.

The problems, values and objectives of the following building blocks are based on stakeholder research carried out in phase 1. The technical specifications (including data and pilot catchments) will be further developed in collaboration with subject-matter experts and data scientists and are subject to change during phase 2 of the project.

Building blocks

Building block 3A: Detecting anomalies in continuous monitoring data

Defined problem and value

- Water companies and regulators are expected to publish continuous monitoring water quality data every hour adhering to the provisional guidance set out by Defra on how to implement section 82 of the Environment Act (Defra, 2023).
- *"Water quality data must be made publicly available in near real time (within one hour), and in a common format across England."* (Defra, 2023)
- This puts additional pressure on water companies, and regulators, to sense-check data quickly before publishing it to the public.
- This building block, and theme, is predominantly relevant for England and English water companies. However, the ability to rapidly quality check data from water quality sensors could benefit other stakeholders, including water companies outside of England who could get more value out of continuous monitoring.
- An AI/ML model that can detect and flag anomalies and faulty outliers quickly, based on a baseline understanding of a given catchment, will speed up the process of adhering to section 82 and reduce the risk of publishing erroneous and misleading data.
- In this context, anomalies refer to any outliers in the data that does not fit within the normal fluctuations and trends in a location. This could both be pollution events, sensor malfunctions, loss of internet connection or other things that might produce an outlier in the data. The purpose of this building block is <u>not</u> to categorise outliers, but simply detect them.

Technical specifications

- **Modelling techniques:** AI/ML model that makes use of historical data to detect and flag anomalies and outliers in incoming continuous monitoring data.
- **Data inputs**
  - Section 82 parameters: ammonia, dissolved oxygen, turbidity, pH and temperature.
  - Additional dynamic input: rainfall data, CSO data.
- **Pilot catchment(s)**
  - River Wensum (10 years of data)
  - Catchments with section 82 trials in Wessex Water and South West Water.
- **Reference(s)**
  - Andrew Johnson, the Rivers Trust, has worked on a proof of concept for this previously.

Output

- Detect and flag anomalies in the data that requires verification.

Use cases:
The real-world application for stakeholders is for the models within this building block to enable:

- Automated outlier detection enables **Regulators** to prioritise inspections at flagged sites, facilitating quicker responses to potential pollution events. The targeted approach improves resource allocation, improved regulatory enforcement, and increased quality and reliability of the data shared with the public.
- Automatic anomaly detection of pollution events of system malfunctions enables **water companies** to conduct rapid corrective actions to protect the environment and/or water quality monitoring infrastructure. Flagging issues before public release of water quality data will improve data quality, streamline compliance and demonstrate transparency, aligning with the Environment Act´s standards
- Environmental charities access to trustworthy near real-time information about the health of their waterbodies and prioritise conservation efforts and focus resources effectively. This capability strengthens the impact of charities by enhancing their data-driven advocacy, resource allocation and engagement with water quality initiatives.

Building block 3B: Categorising anomalies (events or error) in continuous monitoring data

Defined problem and value

- *(This block is a further development of 3A, anomaly detection classification / categorisation)*
- Water companies and regulators are expected to publish continuous monitoring water quality data every hour adhering to section 82 of the Environment Act.
- The next level of the AI/ML model anomaly detection would be the ability to classify and specify the type of anomaly to enable rapid and targeted action. Anomalies such as unexpected changes in the measured parameters can indicate system failures or pollution events that may harm ecosystems and public health.
- Through a time-efficient classification of anomalies, water companies can respond rapidly to issues, minimising environmental damage and avoiding regulatory penalties. Simultaneously, identifying the system defects is important for optimizing system performance and ensuring that the data are reliable and trustworthy for the public.

- For feasibility purposes, a first iteration of this model should only be built to categorise anomalies as either pollution event or error. Later iterations can be developed to include additional categories and sub-categories.

Technical specifications

- **Modelling techniques:** AI/ML model that make use of historical data to categorise anomalies and outliers in incoming continuous monitoring data as either faulty measurements or pollution events.
- **Data inputs**
  - Section 82 parameters: Ammonia, dissolved oxygen, turbidity, pH and temperature.
  - Additional dynamic input: rainfall data, CSO data, reported pollution incidents, pump/treatment works maintenance.
- **Pilot catchment(s)**
  - River Wensum (10 years of data)
  - Catchments with section 82 trials in Wessex Water and South West Water.

Output

- Categorise anomalies in the data as either pollution event or sensor error.

Use cases:

The real-world application for stakeholders is for the models within this building block to enable:

- **Regulators:** atomised classification of anomalies as pollution events of technical errors allows regulatorst to respond more accurately, prioritising pollution events over sensor malfunctions. This enhances regulatory efficiency, reduces unnecessary site visits and ensures compliance actions focus on genuine environmental risks.
- Anomaly classification enables water companies to address issues more effectively. This targeted approach helps companies maintain accurate public and regulatory reporting, improves resource allocation and ensures rapid compliance with environmental standards
- Environmental charities access to trustworthy near real-time information about the health of their waterbodies and prioritise conservation efforts and focus resources effectively. This capability strengthens the impact of charities by enhancing their data-driven advocacy, resource allocation and engagement with water quality initiatives.

Theme 4: Monitoring biodiversity – species and habitats

Theme 4 is the least defined of the four themes being progressed to phase 2 of River Deep Mountain AI. It will only be addressed during phase 2 after theme 1, 2 and 3 and if time allows for it. Details of how building block 4A will be approached and its objective will be defined further in phase 2 if theme 4 is addressed.

The overall objective of this theme is to investigate how AI/ML or remote sensing can support in efforts of monitoring biodiversity in relation to water body health.

Biodiversity is a key indicator of waterbody health. Measurements and changes over time are used a proxy of waterbody health across the sector and can be used as a way of measuring impacts of pollution.

We believe that AI/ML, and in particular computer vision, can support charities, citizen science initiatives and water companies in their efforts to monitor, track and quantify changes in biodiversity over time to get a better understanding of how waterbody health is impacted over time.

"[rivers] create habitat for plants and wildlife, including some of the UK and Ireland's most iconic or rarest species. Water voles, kingfishers, salmon, otters, beavers, and eels all rely on rivers. We're experiencing a biodiversity crisis, so healthy rivers acting as a sanctuary for our native species is vital." (The Rivers Trust, 2024).

Overview

Context – Why is this important

- Biodiversity supports the resilience and stability of ecosystems and plays a key role in water purification, pollination of crops, climate regulation and soil fertility.
- Healthy, diverse ecosystems are more adaptable and better able to withstand disturbances such as disease outbreaks, habitat loss and climate change.
- Biodiversity plays a critical role in maintaining ecosystem balance, helping to regulate species populations and prevent the dominance of invasive species.
- A decline in biodiversity can lead to ecosystem degradation, threatening the resources and services essential for life.
- Current biodiversity monitoring is a combination of fish, invertebrates and plant surveys, but the practises are manual, time consuming and difficult.
- Advancements in AI/ML and computer vision can allow for real-time automated analysis of biodiversity markers and complex ecological data. The utilization of these advanced technologies can help identify species, track

ecological changes and monitor habitat health more accurate and efficient than traditional methods.
- Automated analysis could help validate and make better use of citizen science data.

Stakeholders
- Charities
- Citizen science initiatives
- Water companies

Challenges
- Time-consuming.
- Requires expert knowledge but relies heavily on citizen science.

Opportunities
Our main hypothesis for this theme is that by utilising AI and ML, in particular computer vision, we can support charities, citizen scientists and water companies in their efforts to monitor biodiversity across catchments. Computer vision can be used to speed up current processes of conducting in-person site visits, kick-samples of invertebrates, or other types of on-the-ground monitoring.

"AI or machine learning to identify species in the field that is starting to give us more regularly collectible and serviceable data about biodiversity, it would be really interesting to get into that." (W12, Interviews)

"We did something with a company called AiDash, who did a site-assessment for us using satellite imagery, we were very impressed by the habitat types they could differentiate [...] it had the advantage of being mapped on GIS [...] we are definitely looking at that, as at least a way of speeding up what it is that we would do and when we are engaging with a landowner or a farmer if you could do that first, as in assess the habitat types you then cut down the site visit time to look at condition and anything that might be particular significant [...]." (C7, Interviews)

We have defined one opportunity in this theme where we AI/ML can help improve current practices:

- **Building block 4A:** Using computer vision to monitor and quantify biodiversity.

The problem, value and objective of the following building block are based on stakeholder research carried out in phase 1. The technical specifications (including data and pilot catchments) will be further developed in collaboration with subject-matter experts and data scientists and are subject to change during phase 2 of the project.

Building blocks

Building block 4A: Using computer vision to monitor and quantify biodiversity

Defined problem and value

- Current biodiversity monitoring is a combination of fish, invertebrate and plant surveys, but the practises are manual, time consuming and difficult.
- We believe that by training an image classifier on images indicating biodiversity habitats or specific indicator species, we can develop a model that can help citizen scientists and non-experts determine which species or habitat is present in a given catchment.
- This will support the monitoring and quantification of biodiversity measurements across catchments and improve our overall understanding of waterbody health.
- Several species and habitats are of interest, including macro invertebrates, river flies, woodlands, grasslands, wetlands, etc.
- Problem, value and technical specifications are to be defined more in theme 2 if this building block is addressed.

Technical specifications

- **Model techniques:** Computer vision (photos, earth observation) model
- **Data**:
    - Labelled images: Labelled images of invertebrate species, labelled images of river fly species, labelled satellite images of different habitats, labelled images of indicator species.
- **Data sources:**
    - Citizen science initiatives
    - Smart rivers programme
    - Riverfly census
    - Kick-sample initiatives
- **Reference(s):**
    - INNS (Invasive Non-Native Species) classifiers
    - Flower identification apps.
    - Flora Incognita

Output
Model outputs will be defined more clearly by theme leads during phase 2. Below is suggested preliminary outputs:

- Classifications of species or habitats based on images.

Use cases:
The real-world application for stakeholders is for the models within this building block to enable:

- Regulators to leverage automated identification of indicator species and habitats providing more consistent and scalable biodiversity assessments across water catchments over time. These models can support compliance with biodiversity targets, enabling early detection of environmental risks and identify conservation priorities.
- Water companies to track biodiversity near reservoirs and water treatment sites, allowing them to assess ecological impact of operations and identify areas for improvement. Automated identification and tracking of indicator species will help them conduct faster, cost-effective assessments across multiple sites and provide consistent data for reporting to regulatory bodies.
- Environmental charities to empower citizen science initiatives by making biodiversity monitoring more efficient, accessible and scalable. AI models trained to identify species from photos can help volunteers and communities collect high quality data, enriching biodiversity records and guide conservation efforts at the community level.

3 Next steps

Next steps

As we enter into phase two of the project, each of the four themes will have an assigned 'working group' moving forwards. Each working group will be made up of project partners, TAG members and external stakeholders. Themes will be prioritised as described above (see section 1). During these next project phases, WP2 will continue to ensure that model outputs, accuracy and explanations remain relevant/appropriate for end users and stakeholders. WP2 will continue to conduct stakeholder engagement sessions, helping to facilitate transfer of key knowledge from relevant parties to the data scientists undertaking model development. In addition, WP2 will continue to work with stakeholders to validate the building blocks and their real-world application.

During phase 2, WP3 will be responsible for coordinating all necessary data collection activities. Data requests should be coordinated across-themes, to ensure that data providers are contacted efficiently. WP5 will be responsible for exploring and developing modelling approaches for each of the theme building blocks. Regular feedback should be provided to the rest of the theme working groups, to ensure that appropriate stakeholders and project partners can be utilised effectively during the ongoing development and review process. Finally, during the latter stages of the project (phase 3 onwards), WP7 will further develop the building block use cases, helping define how building blocks can be applied in the 'real-world'.

4 Appendix

The ‘canvas documents’ (deliverable D2.2) for each of the four themes are included for reference below. These canvas documents provide a high-level overview of each theme. In addition, we have included a canvas document for the fifth theme “understanding the impact of interventions on water quality”. Although we are not taking theme five forward in RDMAI (see *Theme Prioritisation* section of report), it is included for documentation and reference purposes. According to our stakeholder engagement insights and prioritisation framework, this theme has the potential to provide a high level of water quality benefit/impact if the modelling is feasible.

Theme 1: Modelling high-priority pollutants.

Theme: Modelling to understand high-priority pollutants

Objective
Overall objective that the opportunities and building blocks are trying to achieve.

- The overall objective of this theme is to apply AI/ML to develop cost-effective and time-efficient ways of modelling high-priority pollutants.
- A key aspect of finding sources of pollution is understanding if, where and when pollution is present in the waterbody.
- Monitoring is expensive, and testing for high-priority pollutants such as *E. coli*, and phosphorous is particularly challenging.
- An approach to overcoming this is to use the data that is already being collected in a better way.

Context of the theme:
Why is this interesting and important.

- Source apportionment modelling is a key driver of investment decisions in the water sector, with tools such as SAGIS playing a central role in catchment management.
- Currently, a SAGIS model is often built and calibrated with limited data (collected bi-weekly, monthly, or annually, and from few locations), leading to inaccurate outputs.
- Patchy data and outputs lead to inaccurate proportions of diffuse and point-source contributions, hindering the effective targeting of interventions.
- This can lead to resources being misallocated, delaying improvements in overall waterbody health.
- Extensive environmental monitoring is not a feasible long-term solution due to high costs and logistical challenges posing significant financial and operational pressure on regulators, water companies and customers.
- Incorporating artificial intelligence (AI) and machine learning (ML) technologies can bridge these gaps by reducing dependency on extensive monitoring and optimising the use of existing data. By enhancing model accuracy and improving the precision of pollution source identification, AI and ML facilitate evidence driven, targeted water quality interventions, ultimately leading to improved waterbody health.

Stakeholders
Who is this theme interesting for and in which context.

- **Water Companies:** Pollution modelling drives investments across the sector. Effective interventions depend on accurate modelling.
- **Environmental Regulators:** Fundamental for fair and effective enforcement of water standards.
- **Local communities:** Essential for public health, biodiversity, recreation & customers' financial burden.

Challenges
Why is it challenging for stakeholders to achieve the objectives related to the theme.

- The current lack of water quality and flow data is one reason why data-reliant SAGIS model outputs can be inaccurate. Filling data gaps is expensive - easily £300,000 to monitor a single parameter in a waterbody (C5, W28, Interviews)
- E.coli and P is high priority for regulators - Monitoring P and E.coli is time consuming and expensive.
- Water companies make significant investments in extensive data collection schemes to challenge model output

"Flow is underrepresented in the Skerne, leading to glaring gaps in SAGIS outputs. Typically, the model will use surrogate river flow data, which is only sense checked every 5 years." (C5, Interviews)

"[...] we actually did a huge catchment monitoring program to improve the quality of the models because the models were showing a really significant impact of the wastewater treatment works and the catchment partnership just said no chance. It's not really them, it's agriculture that's causing the problem here [...] we spent about £350,000 and did quite a lot of detailed sampling right down the river to improve the quality of that model." (W28, Interviews)

"[It] takes c. 1 week to get sample results back from labs, by which time it's too late to work out where pollution came from." (W2, Interviews)

Opportunities
How can the challenges be addressed using AI/ML or remote sensing. Building blocks.

Our main hypothesis for this theme is that by optimising how we model and use the data we already have, we believe that we can improve accuracy and increase understanding of pollutants (sources and impact).

We have defined four opportunities where AI/ML can be applied to overcome some of the specific challenges related to this theme.

"Surrogate for phosphate – would be a gamechanger" (C5, Interviews)

"[How could AI help] link bacteria concentrations to other parameters so that it's possible to measure parameters like E. coli in 'real time'." (W13, Interviews)

In particular we want to address the lack of flow data that is impacting the output of the SAGIS model, as well as the challenge tied to time-consuming lab testing of E. coli and phosphorous that often delays or impedes decisions-making related to those pollutants. Lastly, we want to investigate how well AI/ML can be used to model the travel-time and dilution of pollutants in waterbodies.

This has led to the definition of four potential AI/ML building blocks:

1. **Building block 1A:** Use AI/ML to estimate river flow data at a higher accuracy than current surrogate practices.
2. **Building block 1B:** Use AI/ML to estimate E. coli to reduce reliance on time-consuming lab testing.
3. **Building block 1C:** Use AI/ML to estimate phosphorous to reduce reliance on time-consuming lab testing.
4. **Building block 1D:** Use AI/ML to model high priority pollutants in waterbodies (travel time / decay / die-off)

Level 1: The challenge

High confidence

Building block 1A: Use AI/ML to estimate river flow data at a higher accuracy than current surrogate practices.

- Flow data is limited in most catchments. Flow data is a key parameter going into the SAGIS source apportionment model, and in general essential when trying to understand where pollution is coming from.
- Currently data gaps are filled using surrogate flow data from other similar rivers. These surrogates are sense-checked every 5 years. This approach comes with its own disadvantages.
- To increase the accuracy of SAGIS in individual waterbodies, stakeholders undertake expensive 12-month monitoring programmes of flow data. Monitoring campaigns like this can easily cost upwards of £300,000.
- An AI/ML model that can estimate flow data at a higher accuracy than current surrogate practices will improve current modelling and enable water companies and regulators to get a better, faster and cheaper understanding of where pollution is coming from.

Technical specs X: Data, potential modelling techniques, site-specific or transferable

Modelling techniques: AI/ML/GIS model that considers several layers of data in a catchment to estimate flow based on historical data. Use data that indicates changes in water levels of a river (rain, CSO, effluent), weather and static catchment attributes together with real-time flow data to train a model to estimate flow.

Suggested data inputs: Water coming in, weather, static catchment attributes.
- Historic real-time flow data.
- Water inputs: Rainfall, CSO data / STW flow data, effluent data.
- Weather: Temperature, humidity, and more.
- Static catchment attributes: Mean daily precipitation, elevation, slope, forest, land use, land cover, soil type, soil conductivity and more.

Data sources: Hydrology Data Explorer, MET data, Water Companies, STREAM, Crop Map of England (CROME), Rural Payment Agency, Elevaun, CEH, Soil data.

(Pilot) Catchments: Train on as many catchments as possible in the Hydrology Data Explorer where river flow and rainfall data are available real-time for the past 20 years. Validate and improve in a pilot catchment where new flow data is currently being collected (potentially a catchment in the Thriving Catchments programmes). Could also validate with drone and trial CNSL computer vision stream gauge - Aylem.

Reference(s): Google Research has done something similar to predict extreme floods in ungauged rivers, however their model is not open source, nor available just for flow. Their approach is detailed in medium.com/ai-c-ze/ai

Output X: What type of data (or other output) is coming out of this model and in which format.

Table output: Resembling data output coming from an EA controlled flow monitor / gauge in a table format or similar.

Key parameters: Estimated flow in m3/s, time, date, location.

Model evaluation: Phase 2: Validate against flow collected in catchments outside of the training data. Phase 3: Validate against current surrogate modelling (if possible).

Medium confidence

Building block 1B: Use AI/ML to predict E. coli to reduce reliance on time-consuming lab testing

- Monitoring E. coli is currently constrained by the time it takes to complete a lab test and identify E. Coli colony forming units per 100ml (CFU/100ml). This is a process that, depending on logistics, takes at least 48 hours but also up to a full week.
- E. Coli is a high-priority pollutant for regulators to monitor. It plays a key role in classifying and regulating designated bathing waters.
- In most cases, knowing that bathing water contained E. coli 48 hours ago is not very useful to act or warn.
- An AI/ML model that can use parameters more easily accessible and measurable to predict the concentration of E. Coli in bathing waters will enable water companies to make better decisions and inform local communities on the quality of bathing waters.

Technical specs Y: Data, potential modelling techniques, site-specific or transferable

Modelling techniques: AI/ML/GIS. model that considers several layers of data in a catchment to estimate (and predict) E. coli (CFU).

Suggested data inputs: Water sources, weather, physio-chemical water data, static catchment attributes.
- Lab samples of E. Coli levels.
- Water sources: Rainfall, CSO/STW.
- Weather: Temperature, humidity, more.
- Physio-chemical water data: Ammonia, Dissolved Oxygen, Temperature, pH, BOD, COD, etc.
- Static catchment attributes: Elevation, slope, forest, land use, land cover, livestock, soil type, soil conductivity, salinity, tidal changes and more.

Data sources: Hydrology Data Explorer, MET data, Swimfo/Ed, Water Companies, STREAM, Crop Map of England (CROME), Rural Payment Agency, CEH.

(Pilot) catchments:
- **Inland bathing waters**: Wessex Water (Warleigh Weir) has 2+ years of bacteria data and simple data available. This would most likely result in a site-specific model. - Wessex are expanding the data collection, they will make these datasets available for RDMAI.
- **Estuarine bathing waters**: Alternatively, explore if EA bathing water monitoring data across English estuarine bathing waters can be used to develop a transferable model. Data here is limited to 20 samples a year per location.

Reference(s): Wessex Water and Uniflo has completed trials in Warleigh Weir using a similar approach as above, but without static catchment attributes. Our hope is to use their data to train a new model.

Output Y: What type of data (or other output) is coming out of this model and in which format

Table: With key parameters such as estimated E. coli (cfu), time, date, location.

Model evaluation: Phase 2: Validate against available E. Coli data.

Needs more validation

Building block 1C: Use AI/ML to estimate phosphorous to reduce reliance on time-consuming lab testing.

- Monitoring phosphorous currently requires lab testing and wet chemistry and continuous monitoring is only available using costly sensors.
- Phosphorous is a high-priority pollutant for regulators and water companies to monitor. Phosphorous levels play a key role in the ecological classification of a waterbody.
- An AI/ML model that can use proxy parameters more easily accessible and measurable to estimate the concentration of phosphorous in waterbodies will enable water companies and regulators to get better insight into the impact of phosphorous at a lower cost and higher frequency.

Technical specs Z: Data, potential modelling techniques, site-specific or transferable

Modelling techniques: AI/ML/GIS. model that considers several layers of data in a catchment to estimate phosphorous using proxy parameters.

Suggested data inputs: Focus should be on parameters more common and easily monitorable than phosphorous.
oPhysio-chemical parameters: pH, ammonia, temperature, COD, BOD, dissolved oxygen, etc.

Data sources
- EA, Water companies, STREAM.

(Pilot) catchments: Catchments where phosphorous and a range of other parameters is already being collected. Preferably in real-time. A suggestion could be data from the Ecological Digital Twin in Driffley.

Output Z: What type of data (or other output) is coming out of this model and in which format

Table: With key parameters such as estimated phosphorous, time, date, location.

Needs more validation

Building block 1D: Use AI/ML to model high priority pollutants in waterbodies (travel time / decay / die-off)

- Source detection is challenging due to lack of data points and the complexity of waterbodies leading to ineffective interventions
- Modelling pollutant travel time is a key component for identifying pollutant sources in rivers because it enables the tracing of contaminants from their origin to affected areas
- The movement of pollutants is influenced by factors like flow rates, UV exposure, temperature, and pH which affects their decay and transport.
- Using AI/ML to model pollutant travel time can improve source detection by improving the ability to differentiate between point and diffuse sources
- Our hypothesis is that combining environmental factors with travel dynamics can improve our ability to pinpoint pollution hotspots.

Technical specs Z: Data, potential modelling techniques, site-specific or transferable

Modelling techniques: AI/ML model that considers several layers of data in a catchment to model pollutant travel time.

Suggested data inputs: environmental data including rainfall, temperature, flow rates, conductivity and PH

Data sources
- EA, Water companies, STREAM, weather data

(Pilot) catchments: Catchments where increased wq parameters is already being collected and work is being done to identify sources of pollution and the impact of these, such as in the Thriving catchments initiative (for Phosphorous) and at Warleigh Weir / River Avon, Wessex Water.

Output Z: What type of data (or other output) is coming out of this model and in which format

- GIS layer(s) visualising pollution dynamics within a catchment - spatially and temporally.

Level 2: The building blocks

River Deep Mountain AI

Theme 2: Designing an optimised monitoring plan.

Theme: Designing an optimised monitoring plan

Objective

Overall objective that the opportunities and building blocks are trying to achieve.

- Overall aim of this theme is to use AI/ML models to help reduce the financial costs associated with designing and implementing water quality monitoring schemes.
- The outputs from this theme should help stakeholders to understand the most cost-effective way of holistically monitoring a target catchment. (Instead if monitoring just at CSOs, how does our understanding improve if we use the same number of monitors spaced around catchments).
- Additionally, stakeholders should be able to use the ML/AI model(s) to optimise existing catchment monitoring schemes.

Context of the theme:

Why is this interesting and important.

- Following the recent regulatory changes associated with Section 82 of the Environment Act (2021), water companies will be investing in significant continuous water quality monitoring upstream and downstream of CSO assets. This theme will investigate whether these monitors could provide a more accurate representation of catchment water quality, if a more holistic catchment monitoring approach was used instead - potentially offering a future evidence base to adjust these regulatory standards for the benefit of all stakeholders and waterbody health.
- Current catchment water quality 'datasets' often lack the accuracy and reliability needed to make informed decisions about what and where interventions should be implemented.
- It is often difficult to quantify the accuracy and reliability of existing catchment water quality datasets.
- Designing and implementing additional water quality monitoring schemes to improve the accuracy and reliability of data is expensive and time consuming.
- In many cases, multiple in-person site surveys are required, to help inform where additional monitoring should be undertaken. Even a basic site walk-over can cost "*upwards of £200*" to cover a river reach length of only 500 metres.
- The number of 'monitors' that a stakeholder can install is often limited by cost.

Stakeholders

Who is this theme interesting for and in which context.

- **Water Companies:**
- **Environmental Regulators:**
- **Charities/NGOs:**

Challenges

Why is it challenging for stakeholders to achieve the objectives related to the theme.

- Significant financial costs associated with the design and implementation of water quality monitoring schemes. Including:
  - The cost of preliminary site surveys and design of a monitoring scheme.
  - The capital and operational costs associated with continuous monitoring equipment or sampling schemes.
- Consequently, most catchment monitoring schemes rely on a minimal number of monitoring sites, decreasing the accuracy and reliability of water quality estimations for waterbodies further from from monitoring sites.
- Furthermore, if monitors are located in the 'wrong' location, the usability and quality of data they produce can be severely limited, minimising their cost-effectiveness.
- The accuracy/reliability/uncertainty associated with catchment water quality estimations also tend to be relatively unknown. This makes it challenging to understand where additional monitoring schemes would prove most effective.

Opportunities

How can the challenges be addressed using AI/ML or remote sensing. Building blocks.

- The main aim of this theme is to use AI/ML and remote sensing data to reduce the financial costs associated with designing and implementing water quality monitoring schemes.
- The outputs from this theme should help stakeholders understand the most cost-effective way to monitor water quality within a specified a catchment area.
- Increase understanding about the inaccuracy/uncertainty of existing catchment water quality estimations, helping direct stakeholders towards areas that require additional monitoring programmes.
- Quantify the value of optimising the location and type of sensors within a catchment. Similarly, the value associated with the implementation of additional sensors will be quantified.
- Quantify and compare the value of continuous monitoring points versus sampling programmes. Whilst sampling schemes provide an adequate understanding of gradual changes over time, continuous monitoring excels in providing data during 'irregular' events. Being able to quantify the optimum use for each approach will provide valuable insight when planning catchment monitoring schemes.

This has led to the formulation of four building blocks that will help achieve the above aims:

1. **Building block 2A:** The use of AI/ML and remote sensing to provide a spatial and temporal understanding of potential pollution sources.
2. **Building block 2B:** The use of AI/ML and remote sensing to provide a spatial and temporal understanding of pollution pathways/connectivity/risk.
3. **Building block 2C:** The use of AI/ML to estimate current levels of 'accuracy/reliability/uncertainty' in relation to water quality across a catchment.
4. **Building block 2D:** The use of AI/ML to identify the most optimum monitoring locations within a catchment.

Level 1: The challenge

High confidence

Building block 2A: Use AI/ML and remote sensing to Identify and Map Sources of Pollution

Understanding potential sources of pollution in a catchment currently requires time-consuming on-the-ground walkovers or extensive work with GIS data.

The aim of this building block is to use AI/ML and remote sensing to identify and map pollution sources in a time efficient way. Known sources of pollution should be distinguished from potential sources of pollution.

Illegal and permitted pollution sources should be differentiated clearly. Similarly, the distinction between point source and diffuse source pollution should be highlighted.

Sources of pollution could include: CSOs, WWTW effluents, highway runoff, agricultural runoff, livestock, slurry tanks, septic tanks and RV Sites.

Technical specs X: Data, potential modelling techniques, site-specific or transferable

Modelling techniques: Remote sensing and computer vision (AI/ML) that automates the process of finding potential sources of pollution. For computer vision models, 'papers with code' may be a helpful source of information.

Potential data inputs: satellite imagery, land cover datasets, water company asset data, agricultural asset data, crop cover maps, BGS datasets, EA discharge permits.

- Limitations of remote monitoring data (e.g. cloud cover, frequency of data and size of water body, etc.) should be considered.
- Possible image labeling team needed?

(Pilot) catchments:

Output X: What type of data (or other output) is coming out of this model and in which format.

Output format: GIS layer(s) containing labelled pollution sources.

The building block could be used to help develop a standard approach to desktop surveys. Could this approach be detailed as part of this building block?

Medium confidence

Building block 2B: Use AI/MI and remote sensing to Identify and Map Possible Pollution Pathways/Connectivity

Understanding pollution pathways and connectivity in a catchment currently requires on-the-ground walkovers during rain events.

The aim of this building block is to use AI/ML and remote sensing to identify high risk areas with high connectivity and pollution pathways:

- Pollution pathways should be identified and visualised, both spatially and temporally.
- Assess 'level of risk'.
- Highlight potential pollution 'hot-spots' or 'areas of concern'.
- Possible pathways could include: runoff, river networks, sub-surface pathways, etc.

Technical specs Y: Data, potential modelling techniques, site-specific or transferable

Modelling techniques: Remote sensing and computer vision (AI/ML). May already be mature/existing products available to use.

Potential data inputs: satellite imagery, land cover datasets, soil type, slope, meteorological datasets, etc.

- May be (at least partly) dependent on building block 2A.
- LIDAR (or level data) could prove useful.

(Pilot) catchments: Possibly in Ireland (Daniel Hawtree point of contact) 11 years WQ data (P/N ++) in 6 agricultural sites.

Output Y: What type of data (or other output) is coming out of this model and in which format

Output format: Produce 'connectivity map' of water flow. GIS layer(s) visualising pollution pathways within a catchment - spatially and temporally.

Model output: Try and turn all our data into a connectivity graph (in the mathematical sense) with weights for each parameters and lengths, locations etc

Needs more validation

Building block 2C: Use AI/MI to Estimate levels of 'accuracy/reliability/uncertainty' of water quality across a catchment

- Water quality must first be estimated across the catchment waterbody.
- The level of uncertainty associated with 'current' water quality estimations across the catchment must then be understood and mapped.
- Once this level of uncertainty can be modelled for a catchment, an optimisation process can then be undertaken to reduce the uncertainty as much as possible.
- Uncertainty will be associated with each individual parameter. Although perhaps a total catchment uncertainty value could also be calculated in some manner?
- Will the model aim to understand antecedent/seasonal conditions, or disturbance/pollution events within a catchment? The same model may struggle to do both. Important to account for seasonal variation.

Technical specs Z: Data, potential modelling techniques, site-specific or transferable

Potential modelling techniques:

- Interpolate river parameters along the river network, calculating associated uncertainty as a function of the position.
- The uncertainty along the river network can then be 'summed'/'interpolated' for a total catchment uncertainty value. This could be weighted by importance / risk associated with particular locations.
- Potential model: Bayesian Mixed Effects Model (Australian paper, interpolating and extrapolating temperature). Data needed for this approach.
- Potential model: Physics restricted simulations - e.g. sDNA OCN simulation (Spanish paper).

Potential data sources:

(Pilot) catchments:

- Need a selection of 'data rich' catchments and 'data sparse' catchments. Perhaps 3:1 ratio respectively?

Output Z: What type of data (or other output) is coming out of this model and in which format

Model output: A spatial and temporal representation of water quality across the catchment.

A similar spatial and temporal representation of 'uncertainty' associated with this estimation should be produced.

High confidence

Building block 2D: Use AI/MI to identify the most optimum 'monitoring' locations, time and parameters within a catchment

Placement of monitoring in a catchment has a big impact on the results coming out of a monitoring programme. If monitors are placed poorly the understanding of catchment challenges will also be poor.

The aim of this model is to build a model that recommends the most optimal placement of monitoring within a catchment to get a holistic understanding of the catchment.

- This building block should aim to produce a model that recommends the most optimal placement of 'monitors' within a catchment.
- How many data points are needed to produce a particular level of accuracy within a catchment water quality dataset?
- Additionally, the model should be able to optimise existing monitoring schemes (i.e. not adding more monitors, but adjusting the location of existing monitors).

Technical specs Z: Data, potential modelling techniques, site-specific or transferable

Possible modelling techniques: The sum / 'integrated uncertainty' from building block 2C is calculated for each hypothesis monitoring scheme. Uncertainty should then be minimised (gradient descent).

Potential data inputs: satellite imagery, land cover data, land accessibility data, existing monitoring schemes, tree cover, depths, etc. Local, site-specific factors may be needed.

- Will require additional data collection from Xylem.
- Ideally, real world data from water companies should be used to train and/or validate the model, otherwise stakeholders may be reluctant to trust the outputs.

Output Z: What type of data (or other output) is coming out of this model and in which format

Model output: A clear representation of the most optimum way to monitor a catchment, with a given number of monitors and samples. This should be transferable to a GIS layer.

Documentation: Handbook for how to apply the models for an optimum monitoring campaign.

How might this be applied in real life: For example, an interactive map which initially suggests monitoring and sampling locations. Then if the user were to move a monitoring location, the level of uncertainty within the surrounding river reaches would change as a reflection of that.

Level 2: The building blocks

Theme 3: Getting more value out of continuous monitoring.

Theme: Getting more value out of continuous monitoring

Objective

Overall objective that the opportunities and building blocks are trying to achieve.

- The overall objective of this theme is to use AI/ML models to increase the value and accuracy of the continuous water quality data coming out of section 82 in the Environment Act.
- We believe that AI/ML can be a way of making sure that water companies and regulators get the most value out of this increased monitoring and achieve the overall programme objective of the Environment Act of increasing public understanding.
- The output from this theme should improve stakeholders' ability to sense-check and understand section 82 data in a time-efficient and thorough manner.

Context of the theme:

Why is this interesting and important.

- To better understand the impact of discharges from sewage treatment works on waterbodies the Department of Environment, Food and Rural has put a statutory requirement on English water companies to continuously monitoring upstream and downstream of their assets (Environment Act 2021, Section 82).
- This will increase current OPEX/CAPEX related to monitoring and create unprecedented levels of incoming data for both water companies and regulators.

Stakeholders

Who is this theme interesting for and in which context.

- **Water Companies**
- **Environmental Regulators**
- **Charities/NGOs**

Challenges

Why is it challenging for stakeholders to achieve the objectives related to the theme.

- Increased capital cost of acquiring and deploying sensors, as well as setting up live dashboards, visualisations etc. to adhere to statutory requirements.
- Increased operational cost to maintain sensors.
- Unprecedented amounts of incoming data that must be sense-checked and published within 1 hour every day.
- Together these put water companies and regulators under pressure to handle sensors, data and sense-checking.

Opportunities

How can the challenges be addressed using AI/ML or remote sensing. Building blocks.

- Our main hypothesis for this theme is that by utilising AI and ML we can support water companies in adhering to their statutory requirement of publishing continuous water quality data within 1 hour.
- *"Let's just talk about the data for a minute. We have to publish these results within one hour. On a customer facing dashboard [...] To do that, we need to be confident that the results are accurate, that the results, there's no false flags in there. We need to screen it."* (W3, Interviews)
- [...] *"We can do it with humans or current technology, but definitely the world of AI will play a massive part in achieving that one-hour publication of that data with confidence."* (W3, Interviews)

We have defined two opportunities where AI/ML can be applied to overcome some of the specific challenges related to this theme.

1. **Building block 3A:** Detecting anomalies in continuous monitoring data.
2. **Building block 3B:** Categorising anomalies identified within continuous monitoring data

Level 1: The challenge

High confidence **Building block 3A:** Detecting events & anomalies in continuous monitoring data • There is high public demand for water companies and regulators to make continuous water quality monitoring datasets openly available. Furthermore, water companies in England are expected to publish continuous monitoring water quality data every hour adhering to section 82 of the Environment Act. • *"Water quality data must be made publicly available in near real time (within one hour), and in a common format across England."* (Environment Act, s. 82, 2023) • This puts additional pressure on water companies, and regulators, to sense-check data quickly before publishing it to the public. • An AI/ML model that can detect and flag anomalies and faulty outliers quickly, based on a baseline understanding of a given catchment, will speed up the process of adhering to section 82 and reduce the risk of publishing erroneous and misleading data. Ultimately, ensuring that the continuous water quality data being published achieve the programme objective of increasing public understanding.	**Technical specs X:** Data, potential modelling techniques, site-specific or transferable **Modelling techniques:** **Potential data inputs:** **Potential data sources:** **(Pilot) catchments:** **References:**	**Output X:** What type of data (or other output) is coming out of this model and in which format. **Output format:** Detect and flag anomalies in the data that requires manual/human intervention and verification.

Medium confidence **Building block 3B:** Categorising events & anomalies identified within continuous monitoring data
• Understanding why anomalies have occurred within a continuous monitoring dataset will help provide a greater understanding of water body health within a catchment. If anomalies in continuous monitoring data can be categorised, it will help inform whether (and what type of) further action is needed.
• Model outputs could help inform sensor maintenance schedules. Although this is still a 'reactive' approach, if water companies want to move away from reactive maintenance, then this data could be used to help inform future maintenance schedules or early warnings?
• Anomaly categorisation types may include:
  • Pollution Event (perhaps distinguish between 'known' or 'unknown' pollution events - e.g. known CSO spill data)
  • Sensor 'Malfunction' / Sensor Drift
	Technical specs Y: Data, potential modelling techniques, site-specific or transferable
	Modelling techniques:
	Potential data inputs:
	Potential data sources:
	(Pilot) catchments:
	References:
		Output Y: What type of data (or other output) is coming out of this model and in which format

Building block 3C: n/a or TBC	**Technical specs Z:** Data, potential modelling techniques, site-specific or transferable	**Output Z:** What type of data (or other output) is coming out of this model and in which format
Building block 3D: n/a or TBC	**Technical specs Z:** Data, potential modelling techniques, site-specific or transferable	**Output Z:** What type of data (or other output) is coming out of this model and in which format

Level 2: The building blocks

Theme 4: Monitoring bio-diversity – species and habitat.

Theme: Monitoring impact on biodiversity

Objective
Overall objective that the opportunities and building blocks are trying to achieve.

- The overall objective of this theme is to investigate how AI/ML or remote sensing can support in efforts of monitoring biodiversity in relation to waterbody health.
- Biodiversity is a key indicator of waterbody health. Measurements and changes over time are used a proxy of waterbody health across the sector and can be used as a way of measuring impacts of pollution.
- We believe that AI/ML, and in particular computer vision, can support charities, citizen science initiatives and water companies in their efforts to monitor, track and quantify changes in biodiversity over time to get a better understanding of how waterbody health is impacted over time.
- *"[rivers] create habitat for plants and wildlife, including some of the UK and Ireland's most iconic or rarest species. Water voles, kingfishers, salmon, otters, beavers, and eels all rely on rivers. We're experiencing a biodiversity crisis, so healthy rivers acting as a sanctuary for our native species is vital."* (The Rivers Trust, 2024).

Context of the theme:
Why is this interesting and important.

- Biodiversity supports the resilience and stability of ecosystems and plays a key role in water purification, pollination of crops, climate regulation and soil fertility.
- Healthy, diverse ecosystems are more adaptable and better able to withstand disturbances such as disease outbreaks, habitat loss and climate change.
- Biodiversity plays a critical role in maintaining ecosystem balance, helping to regulate species populations and prevent the dominance of invasive species.
- A decline in biodiversity can lead to ecosystem degradation, threatening the resources and services essential for life.
- Current biodiversity monitoring is a combination of fish, invertebrates and plant surveys, but the practises are manual, time consuming and difficult.
- The advancements in AI/ML and computer vision can allow for real-time automated analysis of biodiversity markers and complex ecological data. The utilization of these advanced technologies can help identify species, track ecological changes and monitor habitat health more accurate and efficient than traditional methods.

Stakeholders
Who is this theme interesting for and in which context.

- **Water Companies**
- **Citizen Science Initiatives**
- **Charities / NGOs**

Challenges
Why is it challenging for stakeholders to achieve the objectives related to the theme.

Opportunities
How can the challenges be addressed using AI/ML or remote sensing. Building blocks.

- *"AI or machine learning to identify species in the field that is starting to give us more regularly collectible and serviceable data about biodiversity, it would be really interesting to get into that."* (W12, Interviews)
- *"We did something with a company called AiDash, who did a site-assessment for us using satellite imagery, we were very impressed by the habitat types they could differentiate [...] it had the advantage of being mapped on GIS [...] we are definitely looking at that, as at least a way of speeding up what it is that we would do and when we are engaging with a landowner or a farmer if you could do that first, as in assess the habitat types you then cut down the site visit time to look at condition and anything that might be particular significant [...]."* (C7, 25:58, Interviews)

Level 1: The challenge

Needs more validation **Building block 4A:** Using computer vision to monitor and quantify biodiversity • Current biodiversity monitoring is a combination of fish, invertebrates and plant surveys, but the practises are manual, time consuming and difficult. • We believe that by training an image classifier on images indicating biodiversity habitats or specific indicator species, we can develop a model that can help citizen scientists and non-experts determine which species or habitat is present in a given catchment. • This will support the monitoring and quantification of biodiversity measurements across catchments and improve our overall understanding of waterbody health. • Several species and habitats are of interest, including macro invertebrate, river flies, woodlands, grasslands, wetlands, etc.	**Technical specs X:** Data, potential modelling techniques, site-specific or transferable **Modelling techniques**: Computer vision (photos, earth observation) model **Potential data inputs:** Labelled images: Labelled images of invertebrate species, labelled images of river fly species, labelled satellite images of different habitats, labelled images of indicator species. **Potential data sources:** • Citizen science initiatives • Smart rivers programme • River fly census • Kick-sample initiatives **(Pilot) catchments:** **References:**	**Output X:** What type of data (or other output) is coming out of this model and in which format. **Output format:** • Classifications of species or habitats based on images.
Building block 4B: n/a or TBC	**Technical specs Y:** Data, potential modelling techniques, site-specific or transferable	**Output Y:** What type of data (or other output) is coming out of this model and in which format
Building block 4C: n/a or TBC	**Technical specs Z:** Data, potential modelling techniques, site-specific or transferable	**Output Z:** What type of data (or other output) is coming out of this model and in which format
Building block 4D: n/a or TBC	**Technical specs Z:** Data, potential modelling techniques, site-specific or transferable	**Output Z:** What type of data (or other output) is coming out of this model and in which format

Level 2: The building blocks

Theme 5 (not continued in RDMAI): Understanding the impact of interventions.

Theme: Understand the impact of interventions to improve waterbody health

Objective

Overall objective that the opportunities and building blocks are trying to achieve.

- The overall objective of this theme is to use AI/ML models to increase understanding of how water quality interventions have impacted waterbody health at a national scale.
- Additionally, by understanding the impact that historical interventions have had on water quality, future investment in WQ interventions can be clearly justified.

Context of the theme:

Why is this interesting and important.

- Due to the sporadic and heterogeneous nature of historical nature based solutions, along with the lack of water quality monitoring post-implementation, it has remained difficult to quantify the effectiveness of these different interventions.
- Consequently, it remains challenging for stakeholders to justify the use of nature based solutions and difficult for regulators to accept them over more traditional 'grey' approaches.
- By understanding the impact that historical nature based solutions have had on water quality, water companies should be able to to better predict the benefits future interventions will bring to a catchment watercourse.
- In addition, by improving sector understanding on the effectiveness of certain interventions, future catchment schemes can ensure the most beneficial intervention option is used within a particular scenario.
- *"[the EA] should have metrics where they say 12 kilometers of river have been improved for example [...] it would be good if we can see or have an additional method of assessing how effective those policy interventions are."* (R14, Interviews)

Stakeholders

Who is this theme interesting for and in which context.

- **Water Companies**
- **Environmental Regulators**
- **Charities/NGOs**

Challenges

Why is it challenging for stakeholders to achieve the objectives related to the theme.

- There is limited water quality data available that has specifically been collected post-implementation of interventions. Where data does exist, it tends to be very site-specific and often isn't collected for a long period of time. Additionally, due to the 'noisy' nature of water quality data, it can be difficult to understand how the a single intervention has actually impacted waterbody health.
- Often water quality interventions are relatively small-scale in comparison to the wider catchment system they are located within. Consequently, the individual impact of a single intervention may be difficult to identify.
- The impact of different NbS's can vary greatly, depending on the implementation method and environmental conditions within that specific site/area.

Opportunities

How can the challenges be addressed using AI/ML or remote sensing. Building blocks.

- Our main hypothesis for this theme is that by utilising AI and ML, we can identify how nature based solutions / WQ interventions have impacted waterbody health.
- Due to the scale of this theme, specific WQ interventions should be prioritised. WQ interventions with the most extensive historical records should be prioritised.
- Correlation between long-term landcover change (i.e. remote sensing) and water quality change, near to sites that have had catchment sensitive farming schemes could add to the analysis.
- Understanding "*how successful a program has been [...]*" and *"[...] being able to attribute certain interventions to environmental improvements, so that we know that customer money that's paying for these upgrades or improvements is actually having this [impact]."* (R14, Interviews)
- *"[...] there will be lots of unknowns about why [an intervention] did or didn't work [...] this is where your tool could fill those gaps [...].* (R14, Interviews)

We have defined X opportunities where AI/ML can be applied to overcome some of the specific challenges related to this theme.

1. **Building block 4A:** Understanding the impact of historical catchment sensitive farming schemes on river water quality.

Level 1: The challenge